

Cloud Computing, Serverless Computing

DS5110: Big Data Systems

Summer 2026

Lecture 6

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UNIVERSITY
of VIRGINIA

Some material taken/derived from:

- Berkeley CS 262a (Spring '18) by Ali Ghodsi and Ion Stoica
- Tyler Harter's HotCloud '18 OpenLambda talk

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Outline

- Cloud computing fundamentals
- Serverless computing and Function-as-a-Service

Cloud computing: Background

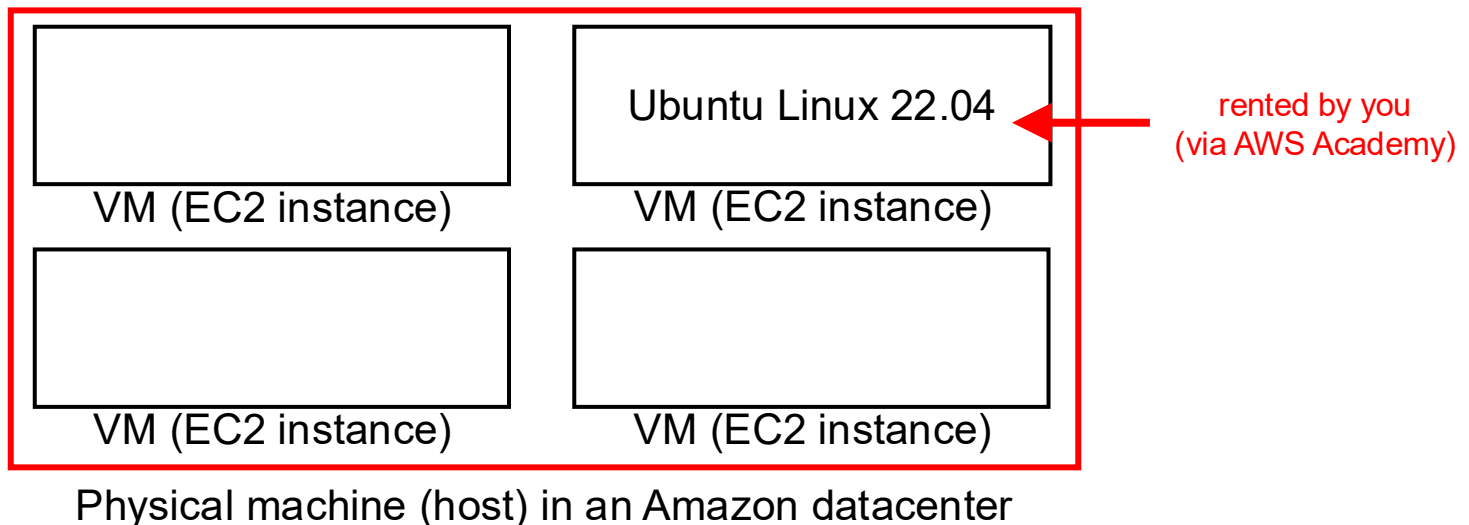
The beginning

“Sometimes you need a lot of processing power; and sometimes you need just a little. Sometimes you need a lot, but you only need it for a limited amount of time.”

-- Jeff Barr (https://aws.amazon.com/blogs/aws/amazon_ec2_beta/)

Amazon Web Services (AWS)

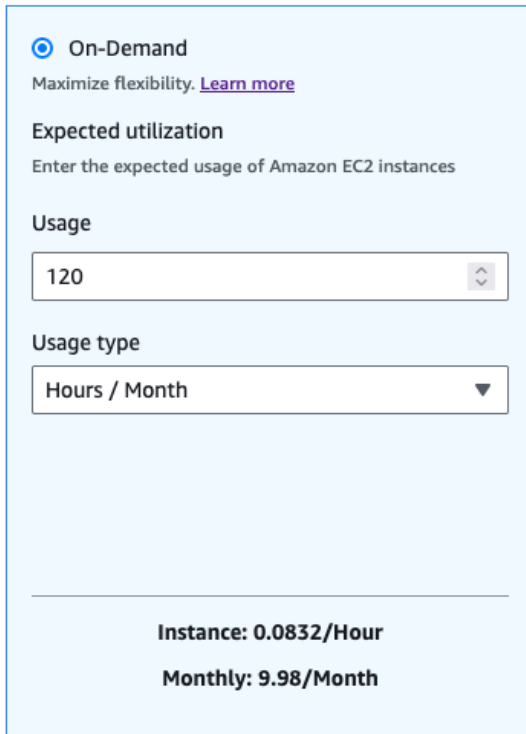
- Elastic Computing Cloud (EC2), rented VMs, launched in 2006
- “**Infrastructure as a Service**” (IaaS): rent infrastructure (compute, storage, network) instead of owning the hardware yourself



VM hours

Pricing summary

t3.large | Family: t3 | 2vCPU | 8 GiB Memory



The screenshot shows the AWS Pricing Calculator interface for an Amazon EC2 On-Demand instance. It includes a section for 'Expected utilization' with a 'Usage' input field set to 120 and a 'Usage type' dropdown set to 'Hours / Month'. At the bottom, it displays the calculated costs: 'Instance: 0.0832/Hour' and 'Monthly: 9.98/Month'.

☒ On-Demand
Maximize flexibility. [Learn more](#)

Expected utilization
Enter the expected usage of Amazon EC2 instances

Usage
120

Usage type
Hours / Month

Instance: 0.0832/Hour
Monthly: 9.98/Month

Amazon EC2 On-Demand instances cost (Monthly): 9.98

Amazon Elastic Block Store (EBS) total cost (Monthly): 1.28

AWS pricing calculator: <https://calculator.aws/#/>

Pricing comparison

- **one VM for a month**: about \$10
- about 120 hours a month (4*30)
- **120 VMs for an hour**: about \$10
- same computation + storage resources
- very different wait time

Be careful!!

- programmers previously optimized when things were **too slow**
- now we need to optimize when it is **too expensive**
- cost is not always obvious at the moment you're running a job (need to do "back of the envelope" estimates before you deploy the resources)

EC2Instances.info

Easy Amazon EC2 Instance Comparison

EC2

RDS

Region: US East (N. Virginia) -

Cost: Hourly -

Reserved: 1-year - No Upfront -

Columns -

Compare Selected

Clear Filters

CSV

Filter: Min Memory (GiB):

0

Min vCPUs:

0

Min Storage (GiB):

0

Name	API Name	Memory	vCPUs	Instance Storage	Network Performance	Linux On Demand cost	Linux Reserved cost	Windows On Demand cost	Windows Reserved cost
Search	Search	Search	Search	Search	Search	Search	Search	Search	Search
M5DN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe SSD	Up to 25 Gigabit	\$0.272000 hourly	\$0.173000 hourly	\$0.456000 hourly	\$0.357000 hourly
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	EBS only	Up to 10 Gigabit	\$0.344000 hourly	\$0.219000 hourly	\$0.712000 hourly	\$0.587000 hourly
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	EBS only	50 Gigabit	\$3.576000 hourly	\$2.253000 hourly	\$5.784000 hourly	\$4.461000 hourly
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe SSD	10 Gigabit	\$0.262000 hourly	\$0.166000 hourly	\$0.446000 hourly	\$0.350000 hourly
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I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NVMe SSD)	100 Gigabit	\$10.848000 hourly	\$7.388000 hourly	\$15.264000 hourly	\$11.804000 hourly
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I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800 GiB SSD	Moderate	\$0.853000 hourly	\$0.424000 hourly	\$0.973000 hourly	\$0.565000 hourly
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T2 Micro	t2.micro	1.0 GiB	1 vCPUs <u>for a 2h 24m burst</u>	EBS only	Low to Moderate	\$0.011600 hourly	\$0.007200 hourly	\$0.016200 hourly	\$0.011800 hourly
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 GiB HDD)	10 Gigabit	\$5.520000 hourly	\$3.216000 hourly	\$6.198000 hourly	\$3.300000 hourly
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Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe SSD	Up to 10 Gigabit	\$1.116000 hourly	\$0.705000 hourly	\$1.668000 hourly	\$1.257000 hourly
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920 GiB SSD	10 Gigabit	\$13.344000 hourly	\$8.223000 hourly	\$16.288000 hourly	\$11.167000 hourly
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	EBS only	100 Gigabit	\$7.152000 hourly	\$4.506000 hourly	\$11.568000 hourly	\$8.922000 hourly
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 GiB SSD)	10 Gigabit	\$6.820000 hourly	\$3.392000 hourly	\$7.782000 hourly	\$4.521000 hourly
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	EBS only	Up to 10 Gigabit	\$1.808000 hourly	\$1.141000 hourly	\$3.280000 hourly	\$2.613000 hourly
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	EBS only	Up to 10 Gigabit	\$0.408000 hourly	\$0.257000 hourly	unavailable	unavailable
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 GiB SSD)	High	\$1.705000 hourly	\$0.848000 hourly	\$1.946000 hourly	\$1.131000 hourly
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NVMe SSD)	Up to 25 Gigabit	\$0.904000 hourly	\$0.616000 hourly	\$1.272000 hourly	\$0.984000 hourly
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	EBS only	Up to 10 Gigabit	\$0.172000 hourly	\$0.109000 hourly	\$0.356000 hourly	\$0.293000 hourly
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	EBS only	Up to 10 Gigabit	\$3.060000 hourly	\$2.088000 hourly	\$3.428000 hourly	\$2.456000 hourly
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs <u>for a 4h 4.8m burst</u>	EBS only	Moderate	\$0.371200 hourly	\$0.230000 hourly	\$0.433200 hourly	\$0.292000 hourly
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 GiB HDD)	10 Gigabit	\$1.872000 hourly	\$1.272000 hourly	\$3.344000 hourly	\$2.744000 hourly
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NVMe SSD)	25 Gigabit	\$6.912000 hourly	\$4.362000 hourly	\$11.328000 hourly	\$8.778000 hourly
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NVMe SSD)	25 Gigabit	\$2.712000 hourly	\$1.847000 hourly	\$3.816000 hourly	\$2.951000 hourly
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	EBS only	10 Gigabit	\$2.128000 hourly	\$1.344000 hourly	\$3.600000 hourly	\$2.816000 hourly
T2 Large	t2.large	8.0 GiB	2 vCPUs <u>for a 7h 12m burst</u>	EBS only	Low to Moderate	\$0.092800 hourly	\$0.057500 hourly	\$0.120800 hourly	\$0.085500 hourly
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920 GiB SSD	High	\$6.669000 hourly	\$4.110000 hourly	\$9.613000 hourly	\$7.054000 hourly
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	EBS only	12 Gigabit	\$2.752000 hourly	\$1.751000 hourly	\$5.696000 hourly	\$4.695000 hourly
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	EBS only	25 Gigabit	\$6.048000 hourly	\$3.810000 hourly	\$10.464000 hourly	\$8.226000 hourly
R5A Large	r5a.large	16.0 GiB	2 vCPUs	EBS only	10 Gigabit	\$0.113000 hourly	\$0.071000 hourly	\$0.205000 hourly	\$0.163000 hourly
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 GiB SSD)	Moderate	\$0.105000 hourly	\$0.073000 hourly	\$0.188000 hourly	\$0.165000 hourly
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	EBS only	20 Gigabit	\$5.424000 hourly	\$3.423000 hourly	\$9.840000 hourly	\$7.839000 hourly
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	EBS only	20 Gigabit	\$4.560000 hourly	\$3.112200 hourly	\$7.504000 hourly	\$6.056200 hourly
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	EBS only	Up to 10 Gigabit	\$0.204000 hourly	\$0.128500 hourly	unavailable	unavailable
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	EBS only	High	\$0.199000 hourly	\$0.126000 hourly	\$0.383000 hourly	\$0.310000 hourly
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480 GiB SSD	Up to 10 Gigabit	\$3.336000 hourly	\$2.056000 hourly	\$4.072000 hourly	\$2.792000 hourly
M5AD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe SSD	Up to 10 Gigabit	\$0.206000 hourly	\$0.132000 hourly	\$0.390000 hourly	\$0.316000 hourly

Other cloud services

- AWS now has > 200 services beyond EC2 (and growing)

Other cloud services

- **IaaS** (Infrastructure as a Service)
 - EC2, other services that feel closer to raw hardware
 - Virtual disks, virtual network, some storage systems, etc.
 - **Cheap + flexible** – you can deploy & run anything on it (Spark, Ray, etc.)
- **PaaS** (Platform as a Service)
 - Cloud providers has deployed systems on the infrastructure; you pay to use the deployed system
 - Databases, application framework/platforms, ML training/deployment systems
 - Less flexible, easier to use
 - Often **more expensive** (though not necessarily more than doing it yourself due to efficiencies available to cloud provider but not you)
- Line between IaaS and PaaS distinction is a bit subjective

Other cloud services

- **FaaS** (Function as a Service)
 - AWS Lambda, the very first FaaS platform across all public cloud providers
 - Users upload code packaged in  “functions” and AWS helps provision it, auto-scale it, and tear it down
 - Finer-grained billing at millisecond level
 - Bundled CPU+memory resources
 - Cheap but not as flexible – you don’t need to worry about deployment

Trends

- What AWS cloud services are most popular today?
- Market share of major cloud providers

Q: How do we know which AWS services are most popular in today's cloud-native apps?

Analyzing AWS' own video series



This is My Architecture

Innovative cloud architectures from AWS partners and customers



Sign in and start building

'This is My Architecture' is a video series that showcases innovative architectural solutions on the AWS Cloud by customers and partners. Each episode examines the most interesting and technically creative elements of each cloud architecture.

[View This is My Architecture special episodes here](#)

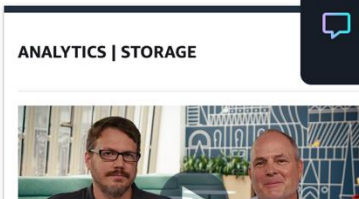
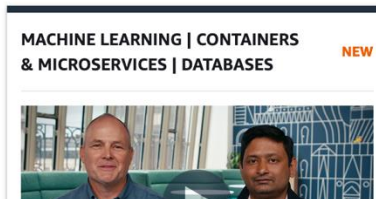
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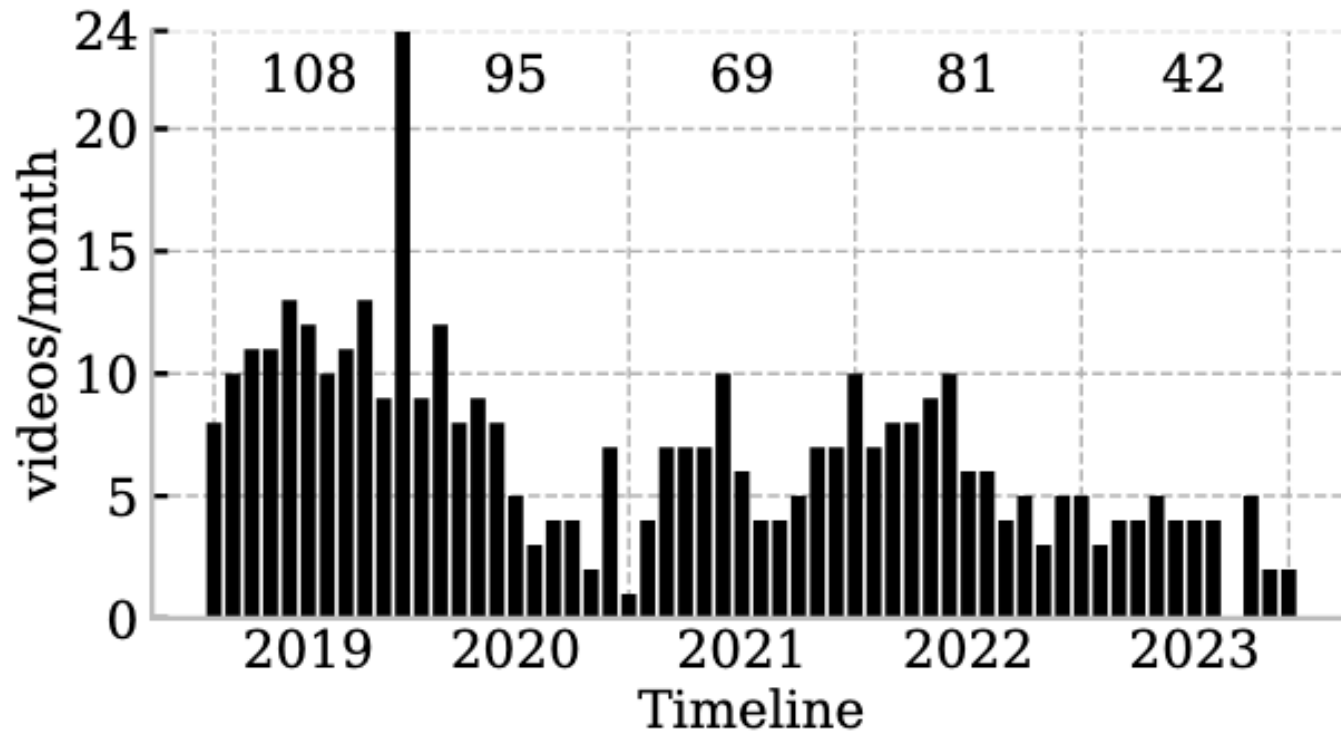
🔍 Search This Is My Architecture Videos

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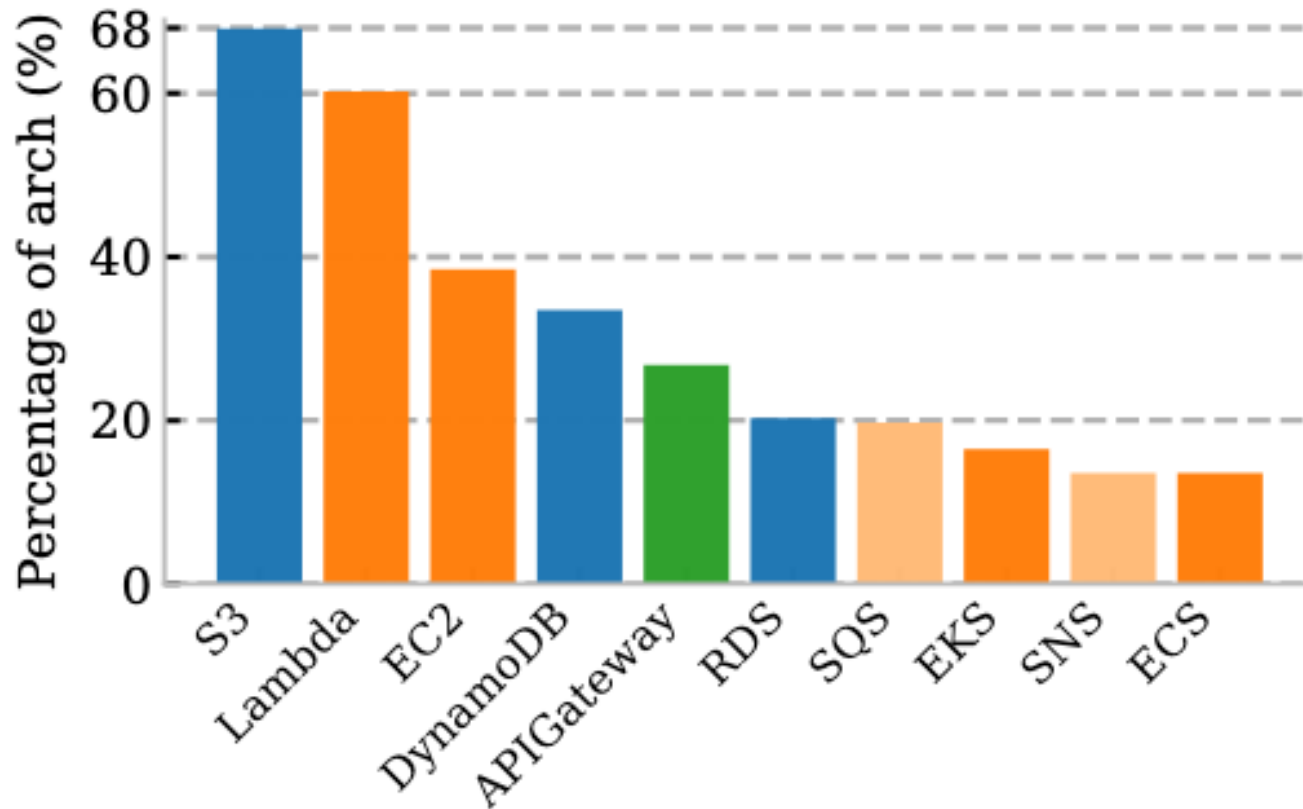
<https://aws.amazon.com/architecture/this-is-my-architecture/>

Distribution of video release date



* Cloudscape: A Study of Storage Services in Modern Cloud Architectures [USENIX FAST 2025]

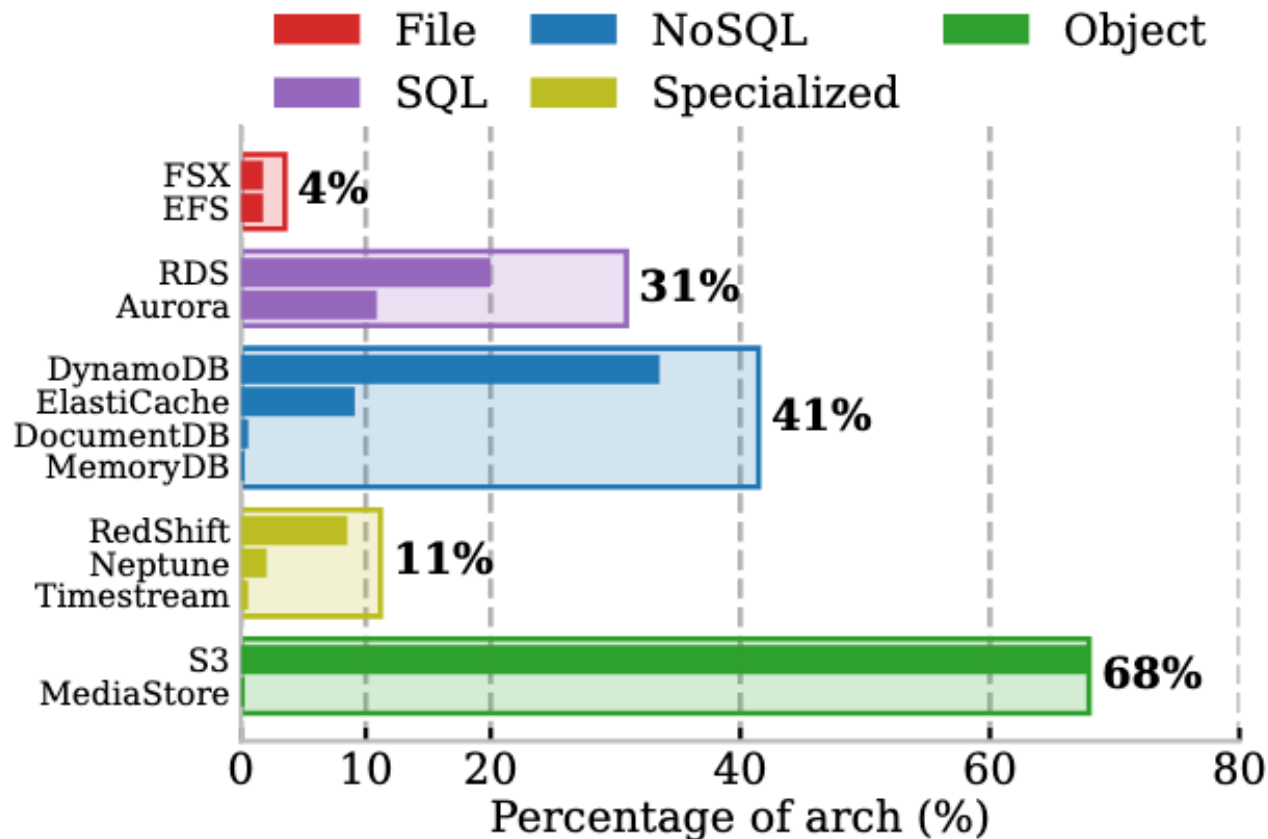
Popularity of different AWS services



All services including compute and storage

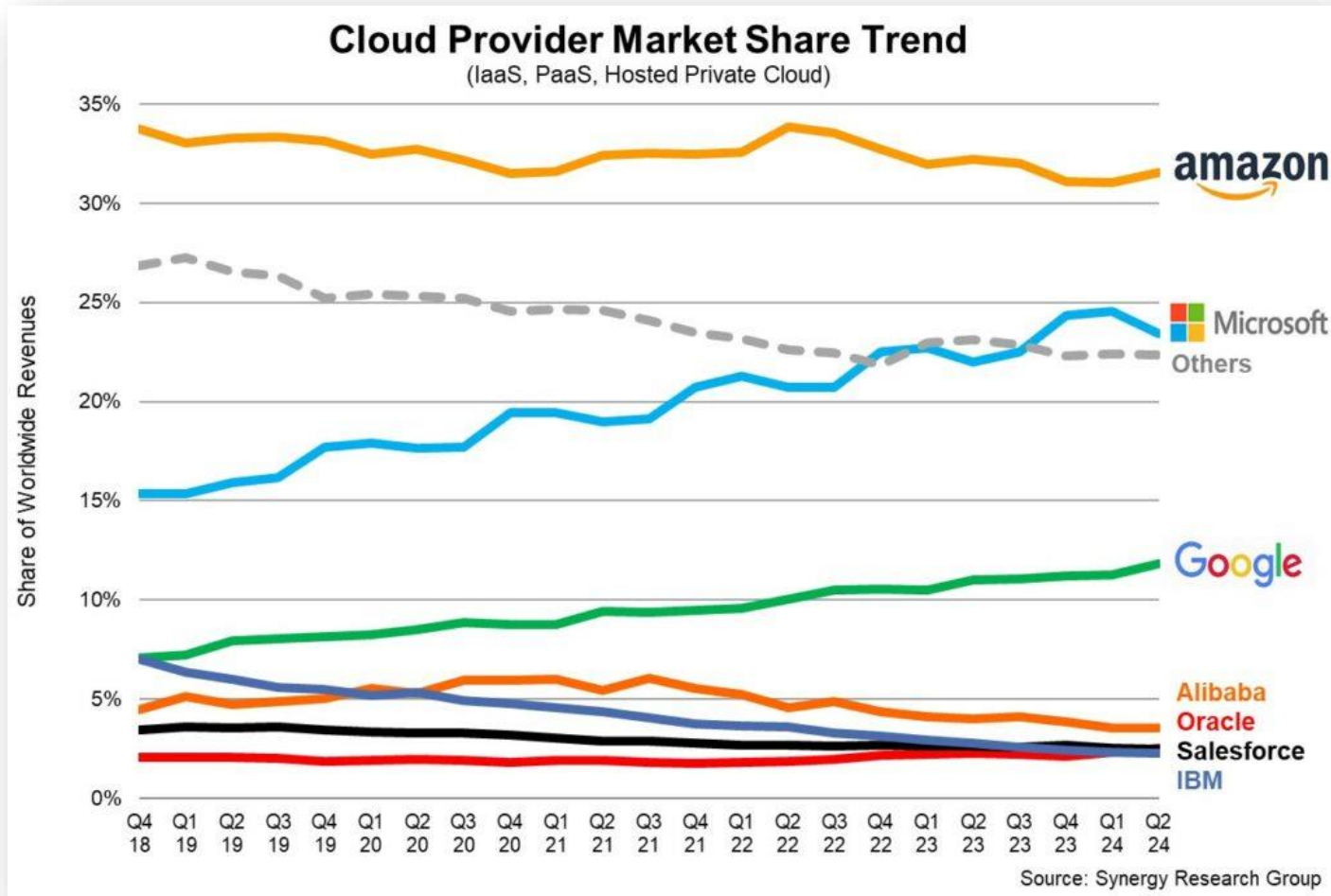
* Cloudscape: A Study of Storage Services in Modern Cloud Architectures [USENIX FAST 2025]

Usage of different storage services



* Cloudscape: A Study of Storage Services in Modern Cloud Architectures [USENIX FAST 2025]

Cloud provider market share trend

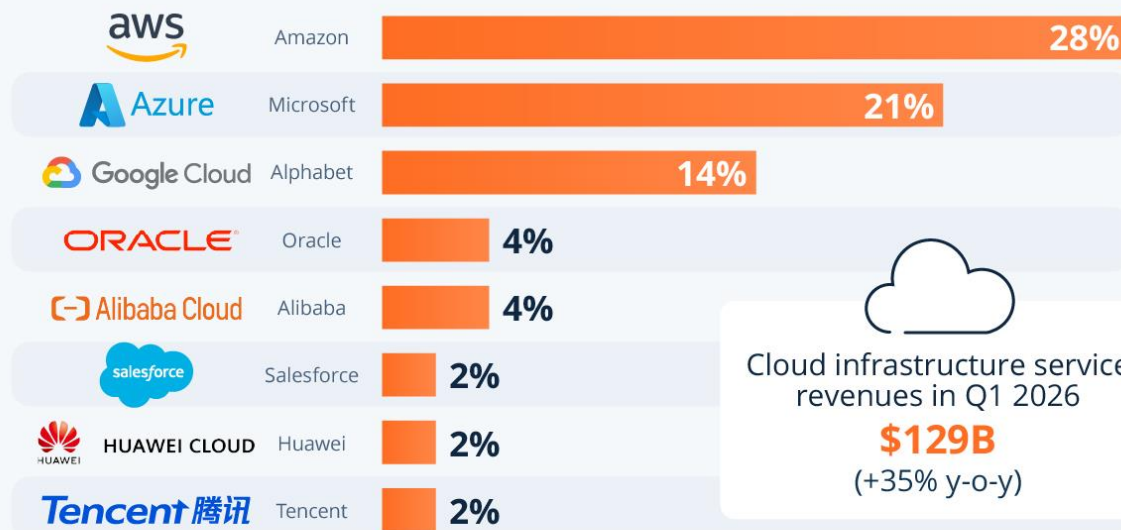


<https://holori.com/cloud-market-share-2024-aws-azure-gcp/>

Cloud provider market share: Q1 26

Big Three Hold Dominant Lead in Accelerating Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q1 2026*



Cloud infrastructure service revenues in Q1 2026
\$129B
(+35% y-o-y)

* Includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group

<https://www.statista.com/chart/18819/worldwide-market-share-of-leading-cloud-infrastructure-service-providers/?srsltid=AfmBOopmp7dSUzGlq-0VA0HNe-QXcWwVI51k6OP7nCqEsEx1qfU8qPek>

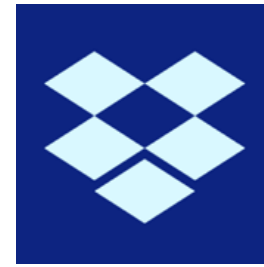


Lock-in

Lock-in

- Customers (tenants) worry: what if the cloud provider increases the price? If it's hard to move to a competing cloud, you're **“locked in”**
- PaaS: services are often unique, and it would be hard to move to a different cloud providers
- IaaS: services like VMs are more uniform – it would be easier to switch to a different cloud to find the cheapest place to rent VMs
- **Data**: cloud providers often **make it free to bring data into the cloud** (ingress) but **expensive to take it out** (egress **\$\$\$\$\$**)

Case study: Dropbox



- A data sync startup founded back in 2008
- Became popular so quickly
 - Peak number of users: 500+ Million
 - Overall amount of data stored: 500 PB
- Initially stored all data on public clouds (AWS)
- Seriously considered to move data out of AWS
- Cloud vendor lock in
 - **Enormous** egress \$\$
- Now still parts of its data services sitting on AWS

Cloud economics and billing models

Tenants: Pay-as-you-go?

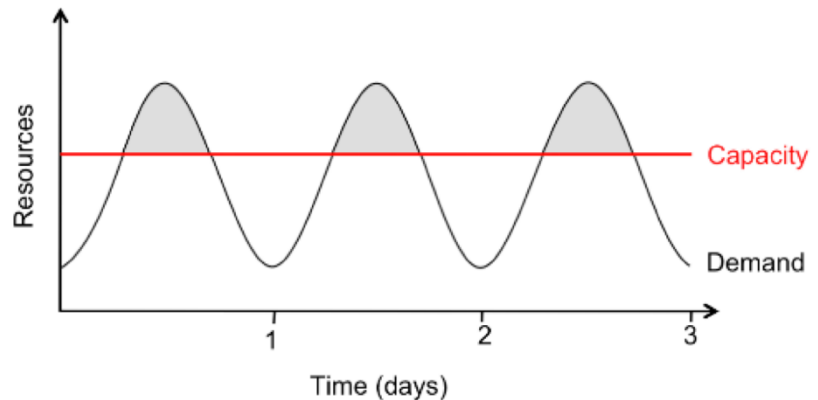
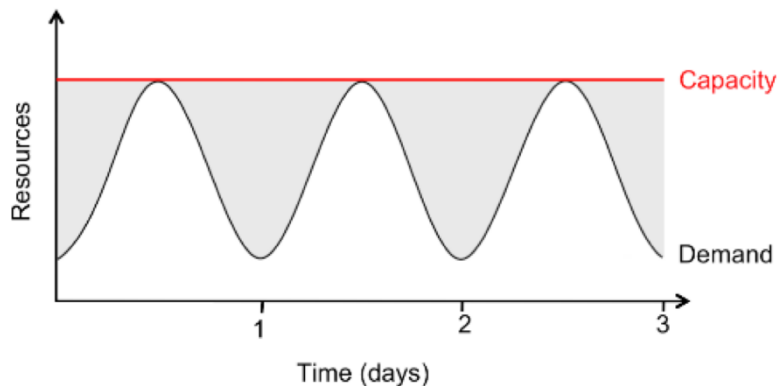
- (**Claimed**) pay-as-you-go pricing
 - Usage-based?
 - Most (compute) services charged per second
 - Except for Lambda, which is charged per millisecond
 - Storage and network services charged per byte
 - No minimum or upfront fee

Tenants: Pay-as-you-go?

- **(Claimed)** pay-as-you-go pricing
 - Usage-based?
 - Most (compute) services charged per second
 - Storage and network services charged per byte
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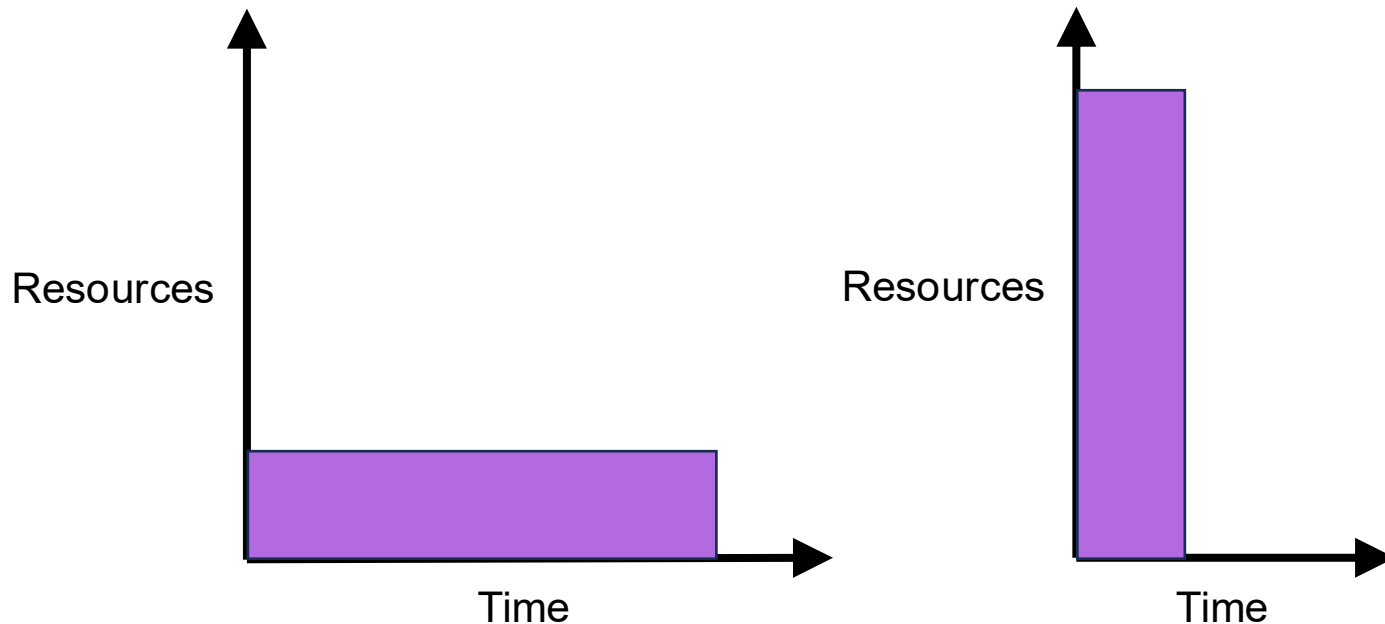
Q: Is the cloud pricing truly pay-as-you-go?

- **Problem:** How to perform strategic planning?



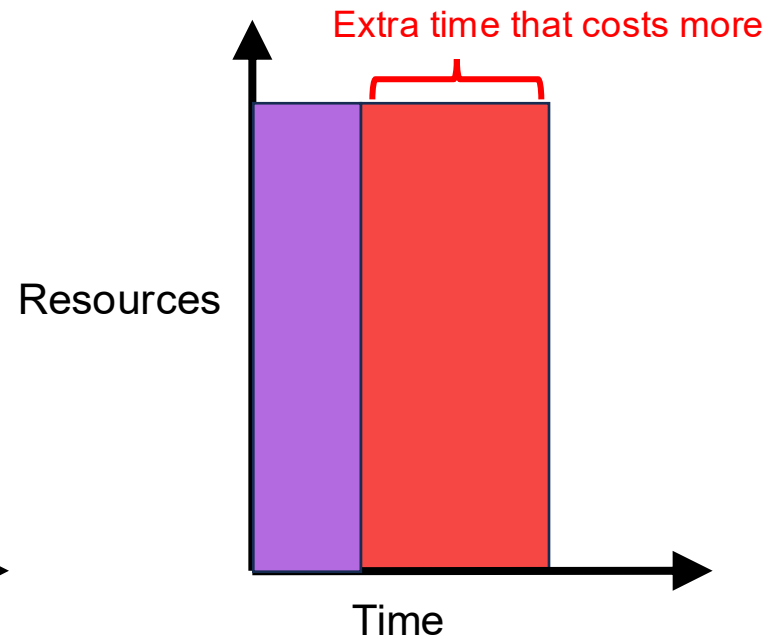
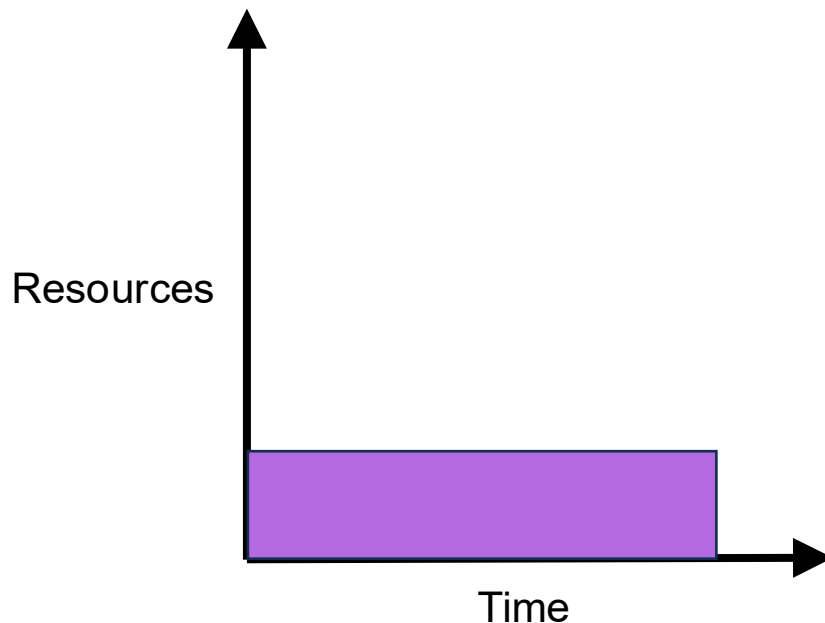
Tenants: Scalability gained?

- (**Ideally**) Linear scalability & perfect elasticity
 - Using 1000 servers for 1 hour costs the same as 1 server for 1000 hours
 - Same price to get a result faster

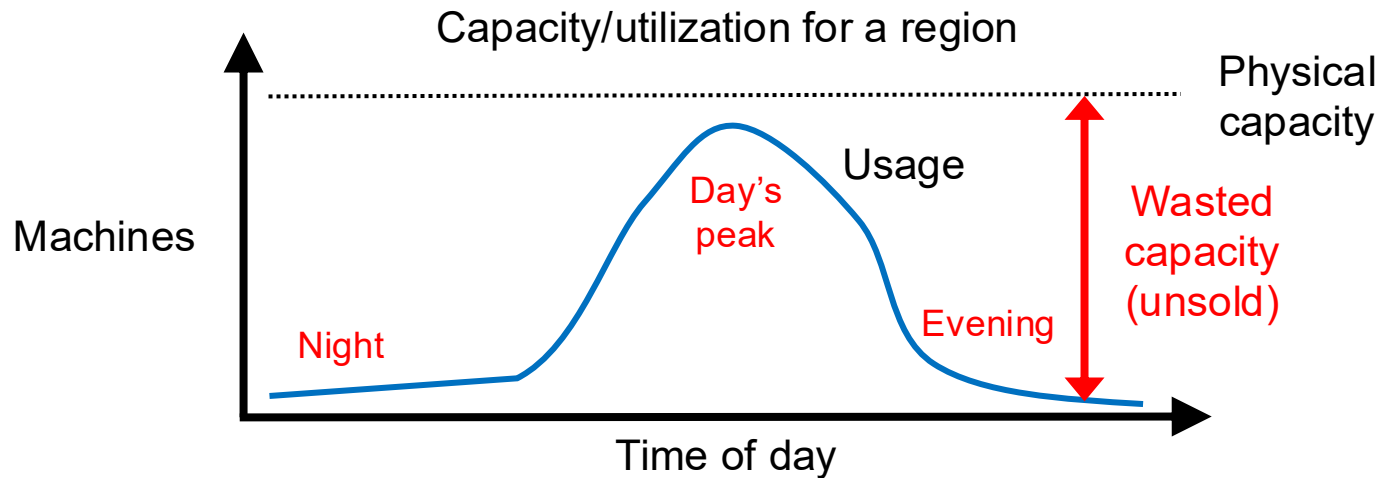


In practice, it really depends, case by case.
Likely the speedup of the computation is much lower than 1000X!

- **(Reality)** Scalability is sublinear and VM scaling is slow.
 - Using 1000 servers for 1+N hour costs **N times** more than 1 server for 1000 hours
 - Often **higher price** to get a result faster

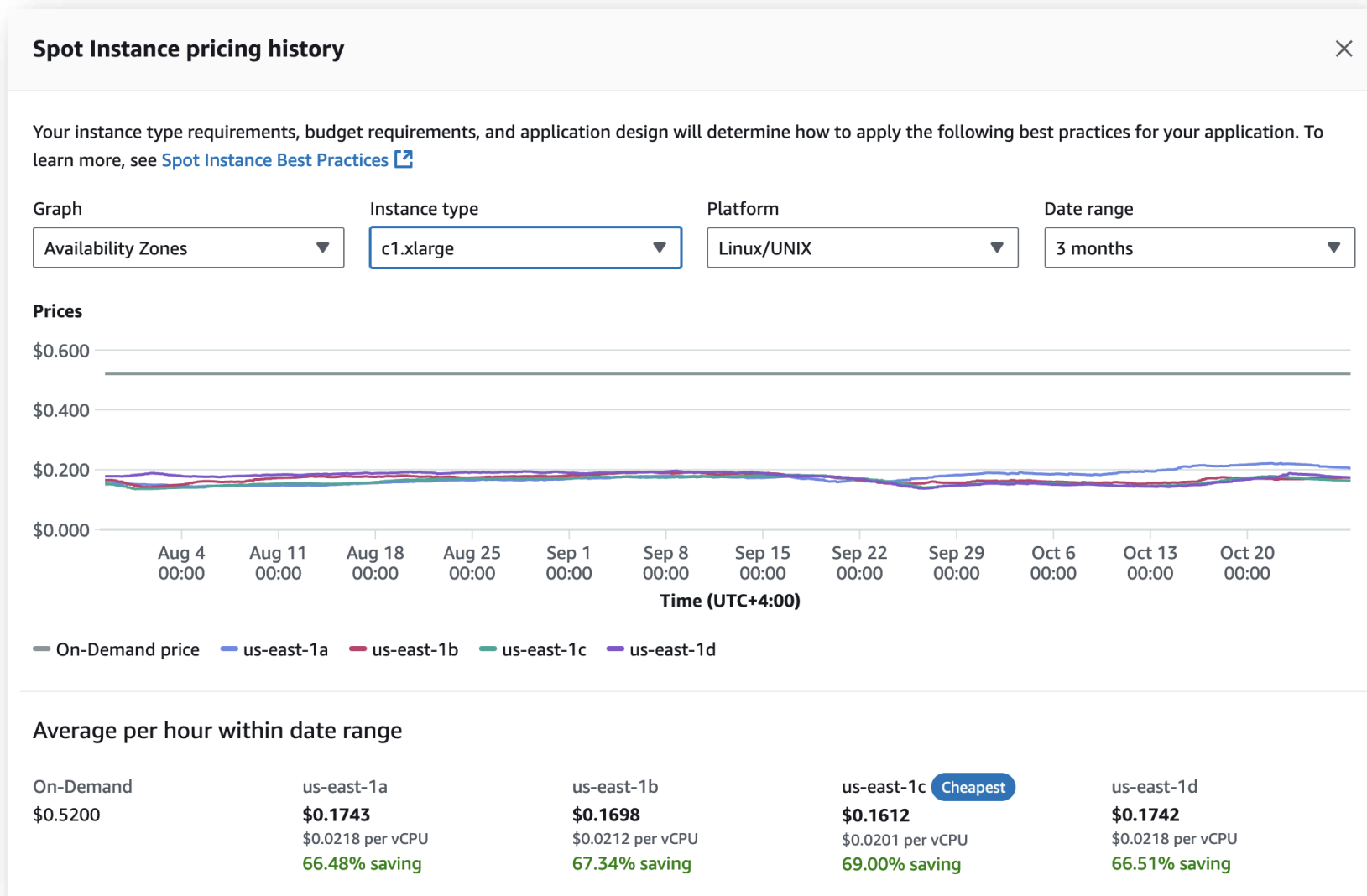


Providers: On-demand vs. spot instances

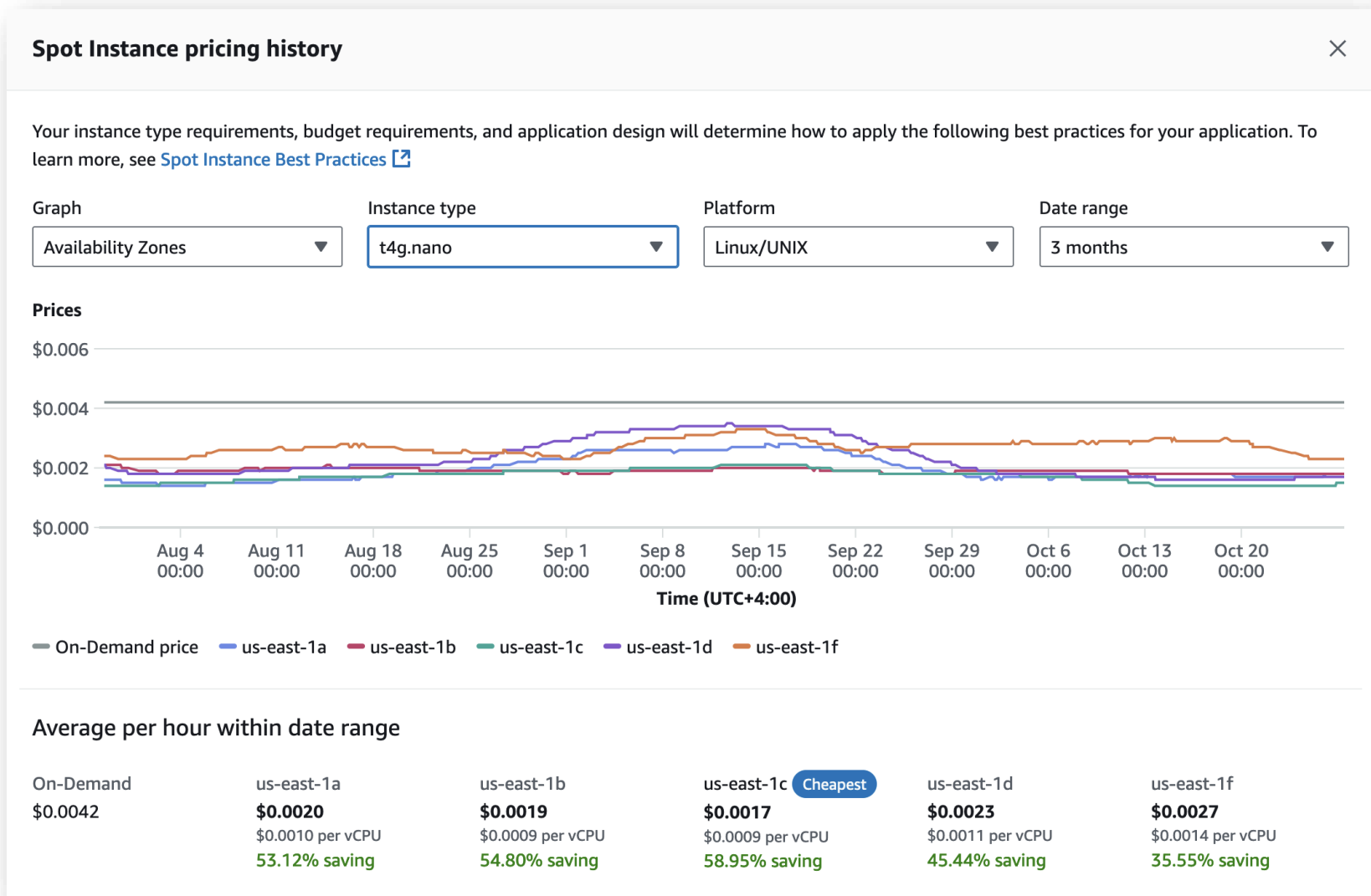


- How to create **incentives** for tenants?
 - Use less at peak time
 - Use more at low times
- Two VM deployment options
 - **On-demand instances**: Constant (high) price. Can generally get a VM. Won't be taken away from you arbitrarily. Used when capacity is needed at specific times.
 - **Spot instances**: Price varies throughout day. If you're not willing to pay enough, your computation waits for a cheaper price. VM might be interrupted ("preempted") once started. Excellent for once-a-day batch jobs.

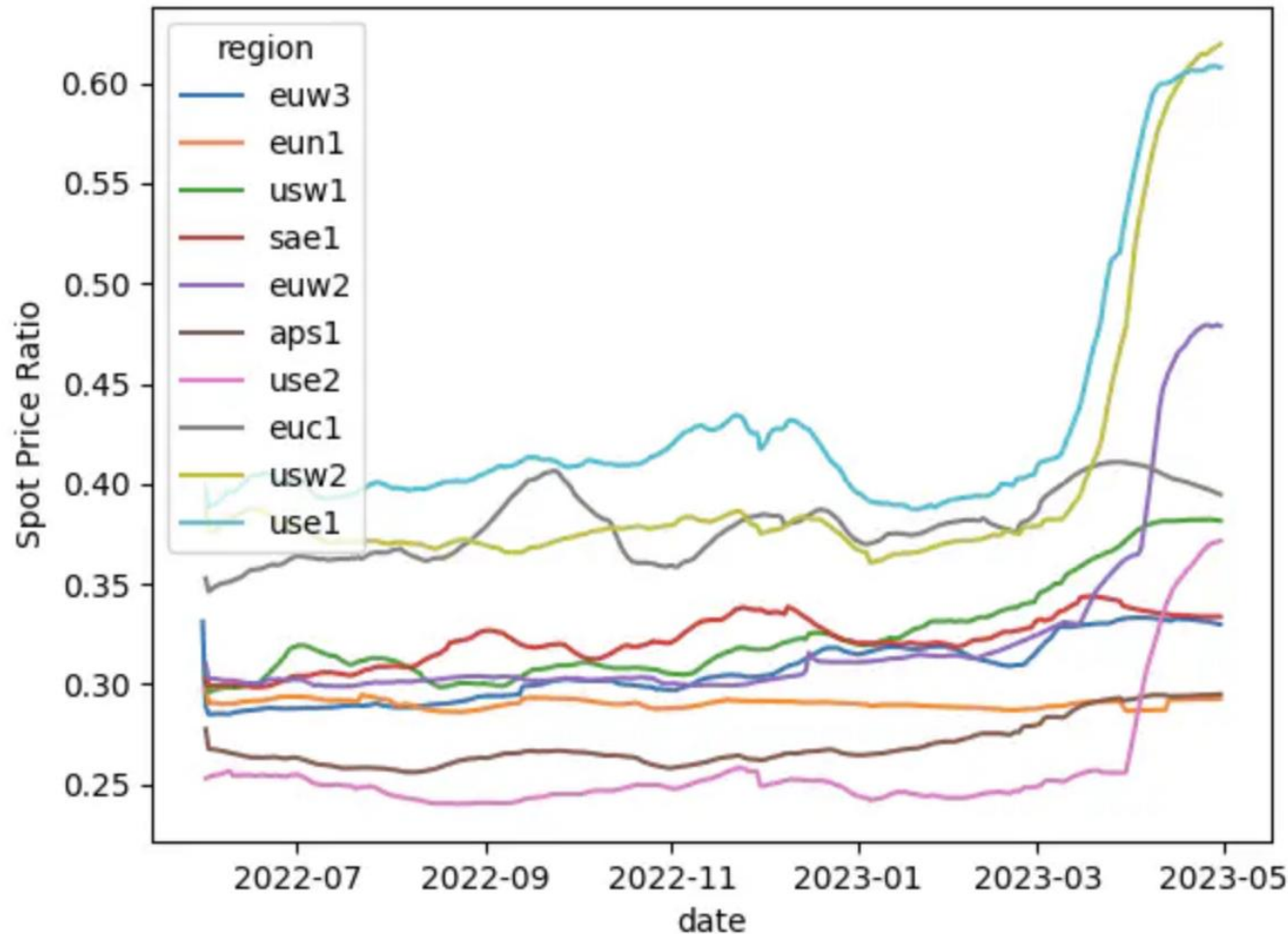
Spot instance pricing (c1.xlarge)



Spot instance pricing (t4g.nano)

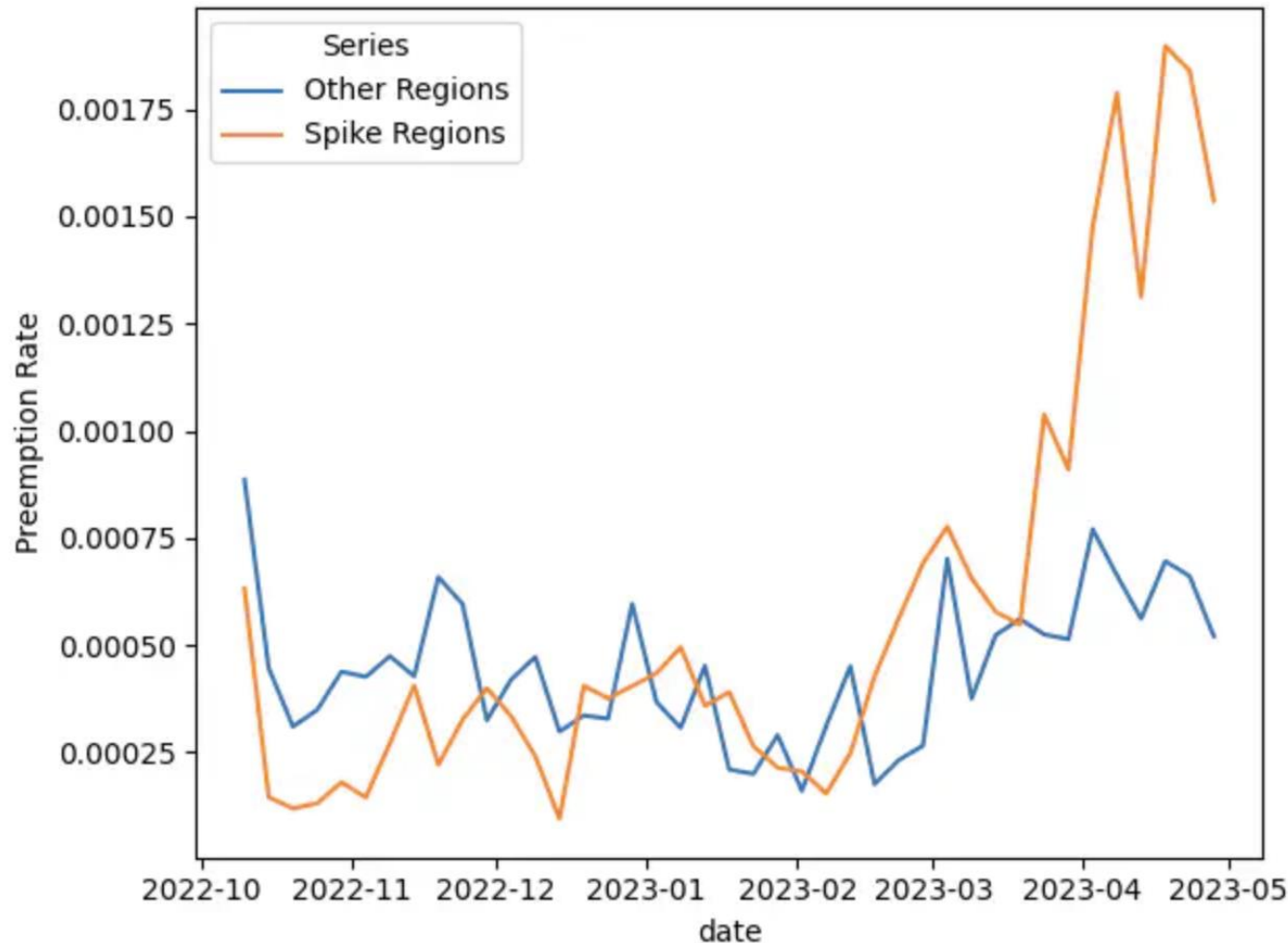


Mean spot price ratios across regions



<https://pauley.me/post/2023/spot-price-trends/>

Spot instance preemption ratio (t3/t4)



<https://pauley.me/post/2023/spot-price-trends/>

Providers: Free tier, discounts at scale

AWS Lambda Pricing

Region: US East (N. Virginia) ↕

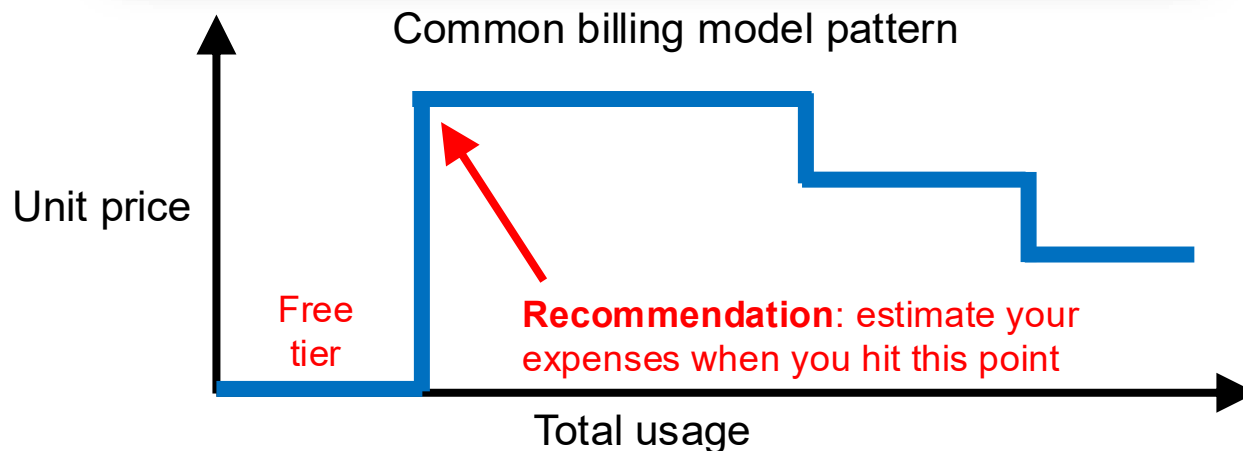
Architecture	Duration
x86 Price	
First 6 Billion GB-seconds / month	\$0.0000166667 for every GB-second
Next 9 Billion GB-seconds / month	\$0.000015 for every GB-second
Over 15 Billion GB-seconds / month	\$0.0000133334 for every GB-second

AWS Lambda example

“The AWS Lambda **free tier** includes one million free requests per month and 400,000 GB-seconds of compute time per month.”

(<https://aws.amazon.com/lambda/pricing/>)

“Duration is calculated from the time your code begins executing until it returns or otherwise terminates, **rounded up to the nearest 1 ms.**”



Recommendation: check if you have a large number of small ops getting rounded up

Outline

- Cloud computing fundamentals
- Serverless computing and Function-as-a-Service

Serverless computing: Motivation

EC2 [RDS](#)

Region: US East (N. Virginia) -

Cost: Hourly -

Reserved: 1-year - No Upfront -

Columns -

Compare Selected

Clear Filters

CSV

Filter: Min Memory (GiB): Min vCPUs: Min Storage (GiB):

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I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NVMe SSD	Up to 25 Gigabit	\$1.356000 hourly	\$0.924000 hourly	\$1.908000 hourly	\$1.476000 hourly
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe SSD	Up to 10 Gigabit	\$1.116000 hourly	\$0.705000 hourly	\$1.668000 hourly	\$1.257000 hourly
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920 GiB SSD	10 Gigabit	\$13.344000 hourly	\$8.223000 hourly	\$16.288000 hourly	\$11.167000 hourly
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	EBS only	100 Gigabit	\$7.152000 hourly	\$4.506000 hourly	\$11.568000 hourly	\$8.922000 hourly
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 GiB SSD)	10 Gigabit	\$6.820000 hourly	\$3.392000 hourly	\$7.782000 hourly	\$4.521000 hourly
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	EBS only	Up to 10 Gigabit	\$1.808000 hourly	\$1.141000 hourly	\$3.280000 hourly	\$2.613000 hourly
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	EBS only	Up to 10 Gigabit	\$0.408000 hourly	\$0.257000 hourly	unavailable	unavailable
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 GiB SSD)	High	\$1.705000 hourly	\$0.848000 hourly	\$1.946000 hourly	\$1.131000 hourly
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NVMe SSD)	Up to 25 Gigabit	\$0.904000 hourly	\$0.616000 hourly	\$1.272000 hourly	\$0.984000 hourly
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	EBS only	Up to 10 Gigabit	\$0.172000 hourly	\$0.109000 hourly	\$0.356000 hourly	\$0.293000 hourly
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	EBS only	Up to 10 Gigabit	\$3.060000 hourly	\$2.088000 hourly	\$3.428000 hourly	\$2.456000 hourly
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs for a 4h 4.8m burst	EBS only	Moderate	\$0.371200 hourly	\$0.230000 hourly	\$0.433200 hourly	\$0.292000 hourly
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 GiB HDD)	10 Gigabit	\$1.872000 hourly	\$1.272000 hourly	\$3.344000 hourly	\$2.744000 hourly
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NVMe SSD)	25 Gigabit	\$6.912000 hourly	\$4.362000 hourly	\$11.328000 hourly	\$8.778000 hourly
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NVMe SSD)	25 Gigabit	\$2.712000 hourly	\$1.847000 hourly	\$3.816000 hourly	\$2.951000 hourly
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	EBS only	10 Gigabit	\$2.128000 hourly	\$1.344000 hourly	\$3.600000 hourly	\$2.816000 hourly
T2 Large	t2.large	8.0 GiB	2 vCPUs for a 7h 12m burst	EBS only	Low to Moderate	\$0.092800 hourly	\$0.057500 hourly	\$0.120800 hourly	\$0.085500 hourly
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920 GiB SSD	High	\$6.669000 hourly	\$4.110000 hourly	\$9.613000 hourly	\$7.054000 hourly
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	EBS only	12 Gigabit	\$2.752000 hourly	\$1.751000 hourly	\$5.696000 hourly	\$4.695000 hourly
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	EBS only	25 Gigabit	\$6.048000 hourly	\$3.810000 hourly	\$10.464000 hourly	\$8.226000 hourly
R5A Large	r5a.large	16.0 GiB	2 vCPUs	EBS only	10 Gigabit	\$0.113000 hourly	\$0.071000 hourly	\$0.205000 hourly	\$0.163000 hourly
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 GiB SSD)	Moderate	\$0.105000 hourly	\$0.073000 hourly	\$0.188000 hourly	\$0.165000 hourly
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	EBS only	20 Gigabit	\$5.424000 hourly	\$3.423000 hourly	\$9.840000 hourly	\$7.839000 hourly
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	EBS only	20 Gigabit	\$4.560000 hourly	\$3.112200 hourly	\$7.504000 hourly	\$6.056200 hourly
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	EBS only	Up to 10 Gigabit	\$0.204000 hourly	\$0.128500 hourly	unavailable	unavailable
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	EBS only	High	\$0.199000 hourly	\$0.126000 hourly	\$0.383000 hourly	\$0.310000 hourly
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480 GiB SSD	Up to 10 Gigabit	\$3.336000 hourly	\$2.056000 hourly	\$4.072000 hourly	\$2.792000 hourly
M5AD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe SSD	Up to 10 Gigabit	\$0.206000 hourly	\$0.132000 hourly	\$0.390000 hourly	\$0.316000 hourly

EC2 Instances (724)

Based on your inputs, this is the lowest-cost EC2 instance: **t4g.nano**

#thecloudistoodamnhard

1. What type of instances?

EC2Instances.info Easy Amazon EC2 Instance Comparison

EC2				
Region: US East (N. Virginia) - Cost: Hourly - Reserved: 1-year - No Upfront - Columns - Compare Selected Clear Filters CSV				
Filter: Min Memory (GiB): 0 Min vCPUs: 0 Min Storage (GiB): 0				
Name	API Name	Memory	vCPUs	Instance Storage
Search	Search	Search	Search	Search
M5DN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	EBS
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	EBS
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	EBS
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NVMe)
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NVMe)
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800 GiB
M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	EBS
T2 Micro	t2.micro	1.0 GiB	1 vCPUs <u>for a 2h 24m burst</u>	EBS
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 GiB NVMe)
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NVMe
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920 GiB
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	EBS
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 GiB)
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	EBS
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	EBS
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 GiB)
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NVMe)
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	EBS
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	EBS
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs <u>for a 4h 4.8m burst</u>	EBS
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 GiB NVMe)
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NVMe)
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NVMe)
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	EBS
T2 Large	t2.large	8.0 GiB	2 vCPUs <u>for a 7h 12m burst</u>	EBS
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920 GiB
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	EBS
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	EBS
R5A Large	r5a.large	16.0 GiB	2 vCPUs	EBS
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 GiB)
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	EBS
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	EBS
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	EBS
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	EBS
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480 GiB
MSAD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe

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1. What type of instances?
2. How many to spin up?

EC2Instances.info Easy Amazon EC2 Instance Comparison

EC2 RDS

Region: US East (N. Virginia) - Cost: Hourly - Reserved: 1-year - No Upfront - Columns - Compare Selected Clear Filters CSV

Filter: Min Memory (GiB): 0 Min vCPUs: 0 Min Storage (GiB): 0

Name	API Name	Memory	vCPUs	Instance Storage
Search	Search	Search	Search	Search
M5DN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NV
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	E
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	E
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	E
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NV
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NV
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800 G
M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	E
T2 Micro	t2.micro	1.0 GiB	1 vCPUs <u>for a 2h 24m burst</u>	E
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 G
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NV
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NV
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	E
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 G
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	E
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	E
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 G
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NV
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	E
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	E
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs <u>for a 4h 4.8m burst</u>	E
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 G
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NV
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NV
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	E
T2 Large	t2.large	8.0 GiB	2 vCPUs <u>for a 7h 12m burst</u>	E
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	E
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	E
R5A Large	r5a.large	16.0 GiB	2 vCPUs	E
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 G
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	E
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	E
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	E
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	E
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480
MSAD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NV

#thecloudistoodamnhard

1. What type of instances?
2. How many to spin up?
3. What base image?

EC2Instances.info Easy Amazon EC2 Instance Comparison

EC2 RDS

Region: US East (N. Virginia) - Cost: Hourly - Reserved: 1-year - No Upfront - Columns - Compare Selected Clear Filters CSV

Filter: Min Memory (GiB): 0 Min vCPUs: 0 Min Storage (GiB): 0

Name	API Name	Memory	vCPUs	Instance Storage
Search	Search	Search	Search	Search
M5DN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	EBS
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	EBS
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	EBS
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NVMe)
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NVMe)
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800 GiB
M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	EBS
T2 Micro	t2.micro	1.0 GiB	1 vCPUs for a 2h 4m burst	EBS
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 GiB NVMe)
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NVMe
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920 GiB
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	EBS
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 GiB)
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	EBS
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	EBS
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 GiB)
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NVMe)
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	EBS
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	EBS
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs for a 4h 4.8m burst	EBS
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 GiB NVMe)
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NVMe)
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NVMe)
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	EBS
T2 Large	t2.large	8.0 GiB	2 vCPUs for a 7h 12m burst	EBS
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920 GiB
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	EBS
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	EBS
R5A Large	r5a.large	16.0 GiB	2 vCPUs	EBS
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 GiB)
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	EBS
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	EBS
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	EBS
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	EBS
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480 GiB
MSAD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe

#thecloudistoodamnhard

1. What type of instances?
2. How many to spin up?
3. What base image?
4. On-demand or spot?

EC2Instances.info Easy Amazon EC2 Instance Comparison

EC2 RDS

Region: US East (N. Virginia) - Cost: Hourly - Reserved: 1-year - No Upfront - Columns - Compare Selected Clear Filters CSV

Filter: Min Memory (GiB): 0 Min vCPUs: 0 Min Storage (GiB): 0

Name	API Name	Memory	vCPUs	Instance Storage
Search	Search	Search	Search	Search
M5DN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	EBS
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	EBS
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	EBS
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NVMe)
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NVMe)
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NVMe
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800 GiB
M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	EBS
T2 Micro	t2.micro	1.0 GiB	1 vCPUs for a 2h 4m burst	EBS
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 GiB NVMe)
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NVMe
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920 GiB
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	EBS
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 GiB)
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	EBS
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	EBS
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 GiB)
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NVMe)
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	EBS
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	EBS
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs for a 4h 4.8m burst	EBS
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 GiB NVMe)
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NVMe)
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NVMe)
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	EBS
T2 Large	t2.large	8.0 GiB	2 vCPUs for a 7h 12m burst	EBS
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920 GiB
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	EBS
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	EBS
R5A Large	r5a.large	16.0 GiB	2 vCPUs	EBS
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 GiB)
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	EBS
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	EBS
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	EBS
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	EBS
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480 GiB
MSAD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NVMe

#thecloudistoodamnhard

1. What type of instances?
2. How many to spin up?
3. What base image?
4. On-demand or spot?
5. What storage service to use?

EC2Instances.info Easy Amazon EC2 Instance Comparison

EC2 RDS

Region: US East (N. Virginia) - Cost: Hourly - Reserved: 1-year - No Upfront - Columns - Compare Selected Clear Filters CSV

Filter: Min Memory (GiB): 0 Min vCPUs: 0 Min Storage (GiB): 0

Name	API Name	Memory	vCPUs	Instance Storage
Search	Search	Search	Search	Search
MSDN Extra Large	m5dn.xlarge	16.0 GiB	4 vCPUs	150 GiB NV
M5A Double Extra Large	m5a.2xlarge	32.0 GiB	8 vCPUs	E
R5N 12xlarge	r5n.12xlarge	384.0 GiB	48 vCPUs	E
R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	E
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NV
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NV
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800
MSN 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	E
T2 Micro	t2.micro	1.0 GiB	1 vCPUs for a 2h 4m burst	E
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 G
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NV
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NV
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920
R5N 24xlarge	r5n.24xlarge	768.0 GiB	96 vCPUs	E
I2 Eight Extra Large	i2.8xlarge	244.0 GiB	32 vCPUs	6400 GiB (8 * 800 G
R5A Eight Extra Large	r5a.8xlarge	256.0 GiB	32 vCPUs	E
A1 Metal	a1.metal	32.0 GiB	16 vCPUs	E
I2 Double Extra Large	i2.2xlarge	61.0 GiB	8 vCPUs	1600 GiB (2 * 800 G
I3EN Double Extra Large	i3en.2xlarge	64.0 GiB	8 vCPUs	5000 GiB (2 * 2500 GiB NV
M5A Extra Large	m5a.xlarge	16.0 GiB	4 vCPUs	E
P3 Double Extra Large	p3.2xlarge	61.0 GiB	8 vCPUs	E
T2 Double Extra Large	t2.2xlarge	32.0 GiB	8 vCPUs for a 4h 4.8m burst	E
H1 Eight Extra Large	h1.8xlarge	128.0 GiB	32 vCPUs	8000 GiB (4 * 2000 G
R5D 24xlarge	r5d.24xlarge	768.0 GiB	96 vCPUs	3600 GiB (4 * 900 GiB NV
I3EN 6xlarge	i3en.6xlarge	192.0 GiB	24 vCPUs	15000 GiB (2 * 7500 GiB NV
R4 High-Memory Eight Extra Large	r4.8xlarge	244.0 GiB	32 vCPUs	E
T2 Large	t2.large	8.0 GiB	2 vCPUs for a 7h 12m burst	E
X1 Extra High-Memory 16xlarge	x1.16xlarge	976.0 GiB	64 vCPUs	1920
M5A 16xlarge	m5a.16xlarge	256.0 GiB	64 vCPUs	E
R5 Metal	r5.metal	768.0 GiB	96 vCPUs	E
R5A Large	r5a.large	16.0 GiB	2 vCPUs	E
C3 High-CPU Large	c3.large	3.75 GiB	2 vCPUs	32 GiB (2 * 16 G
R5A 24xlarge	r5a.24xlarge	768.0 GiB	96 vCPUs	E
G3 16xlarge	g3.16xlarge	488.0 GiB	64 vCPUs	E
A1 Double Extra Large	a1.2xlarge	16.0 GiB	8 vCPUs	E
C4 High-CPU Extra Large	c4.xlarge	7.5 GiB	4 vCPUs	E
X1E Quadruple Extra Large	x1e.4xlarge	488.0 GiB	16 vCPUs	480
MSAD Extra Large	m5ad.xlarge	16.0 GiB	4 vCPUs	150 GiB NV

#thecloudistoodamnhard

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R5AD Extra Large	r5ad.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
R5N Extra Large	r5n.xlarge	32.0 GiB	4 vCPUs	E
I3EN 12xlarge	i3en.12xlarge	384.0 GiB	48 vCPUs	30000 GiB (4 * 7500 GiB NV
I3EN Metal	i3en.metal	768.0 GiB	96 vCPUs	60000 GiB (8 * 7500 GiB NV
R5DN Extra Large	r5dn.xlarge	32.0 GiB	4 vCPUs	150 GiB NV
I2 Extra Large	i2.xlarge	30.5 GiB	4 vCPUs	800
M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	E
T2 Micro	t2.micro	1.0 GiB	1 vCPUs for a 2h 4m burst	E
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 G
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NV
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NV
X1E 16xlarge	x1e.16xlarge	1952.0 GiB	64 vCPUs	1920
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M5N 16xlarge	m5n.16xlarge	256.0 GiB	64 vCPUs	EBS
T2 Micro	t2.micro	1.0 GiB	1 vCPUs for a 2h 24m burst	EBS
D2 Eight Extra Large	d2.8xlarge	244.0 GiB	36 vCPUs	48000 GiB (24 * 2000 GiB NVMe)
I3EN 3xlarge	i3en.3xlarge	96.0 GiB	12 vCPUs	7500 GiB NVMe
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	12 vCPUs	450 GiB NVMe
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#thecloudistoodamnhard

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8. Go back to Step 1...

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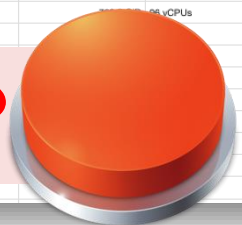
#thecloudistoodamn

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R5 Metal	r5.metal		32 vCPUs	
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Why is there no “cloud button”?



~~Decision paralysis??~~ Go for Serverless!



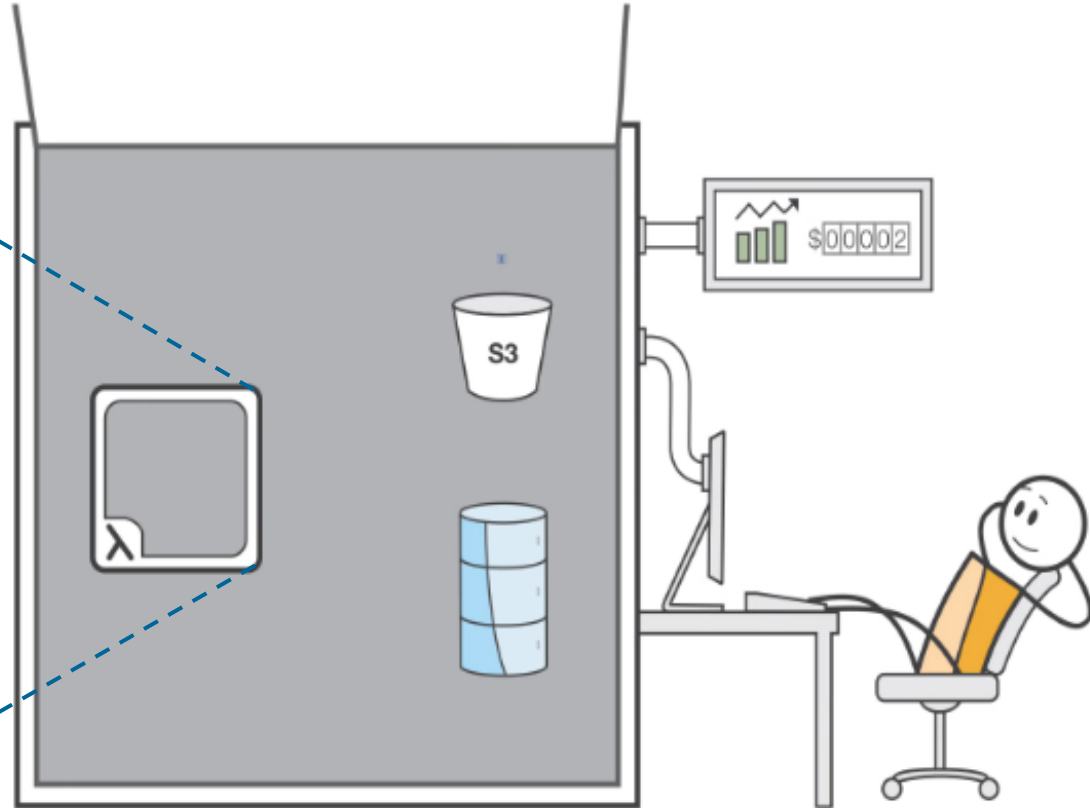
Amazon
Lambda



Google Cloud Functions



Microsoft Azure Functions

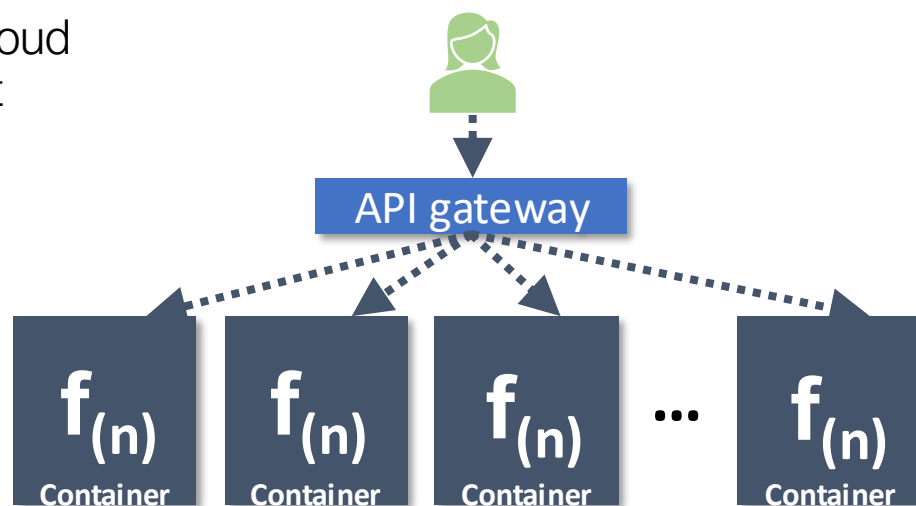


What is serverless computing?

What is serverless computing?

Serverless computing (Function-as-a-Service, or FaaS) is a **programming abstraction** that enables users to upload programs, run them at **(virtually)** any scale, and **pay only for the resources used**

- **Function-as-a-Service (FaaS):** Cloud functions as a basic deployment unit



Function-as-a-Service (FaaS)

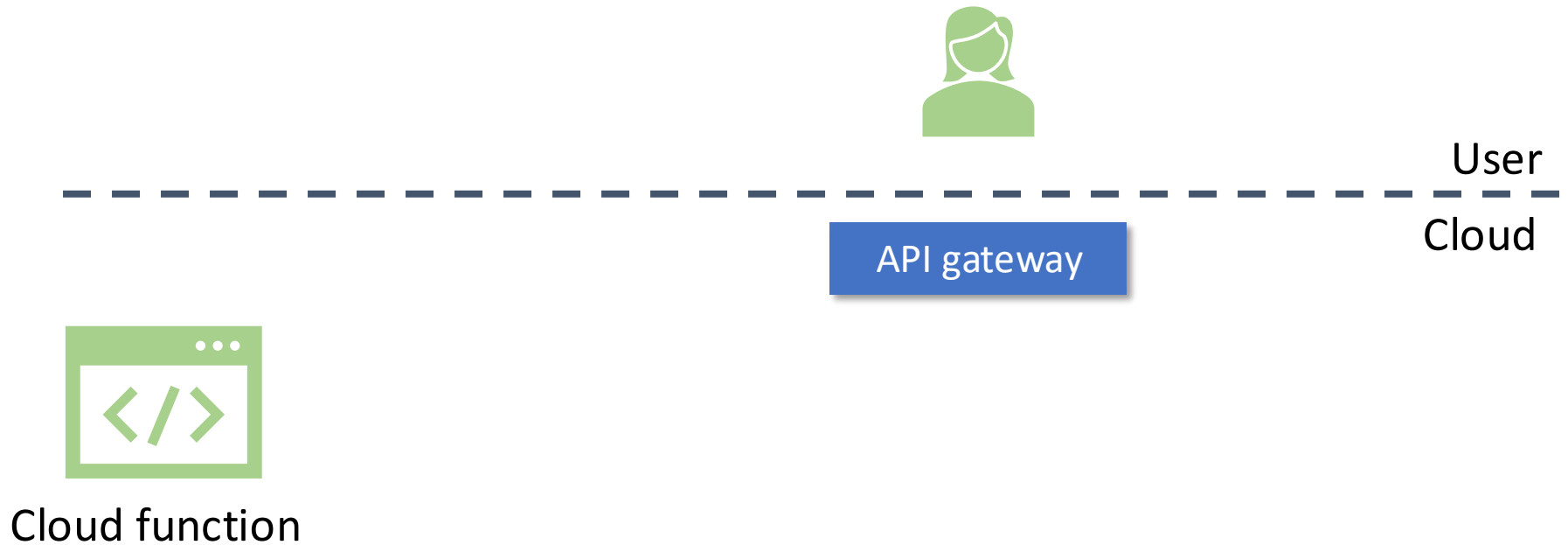


User
Cloud

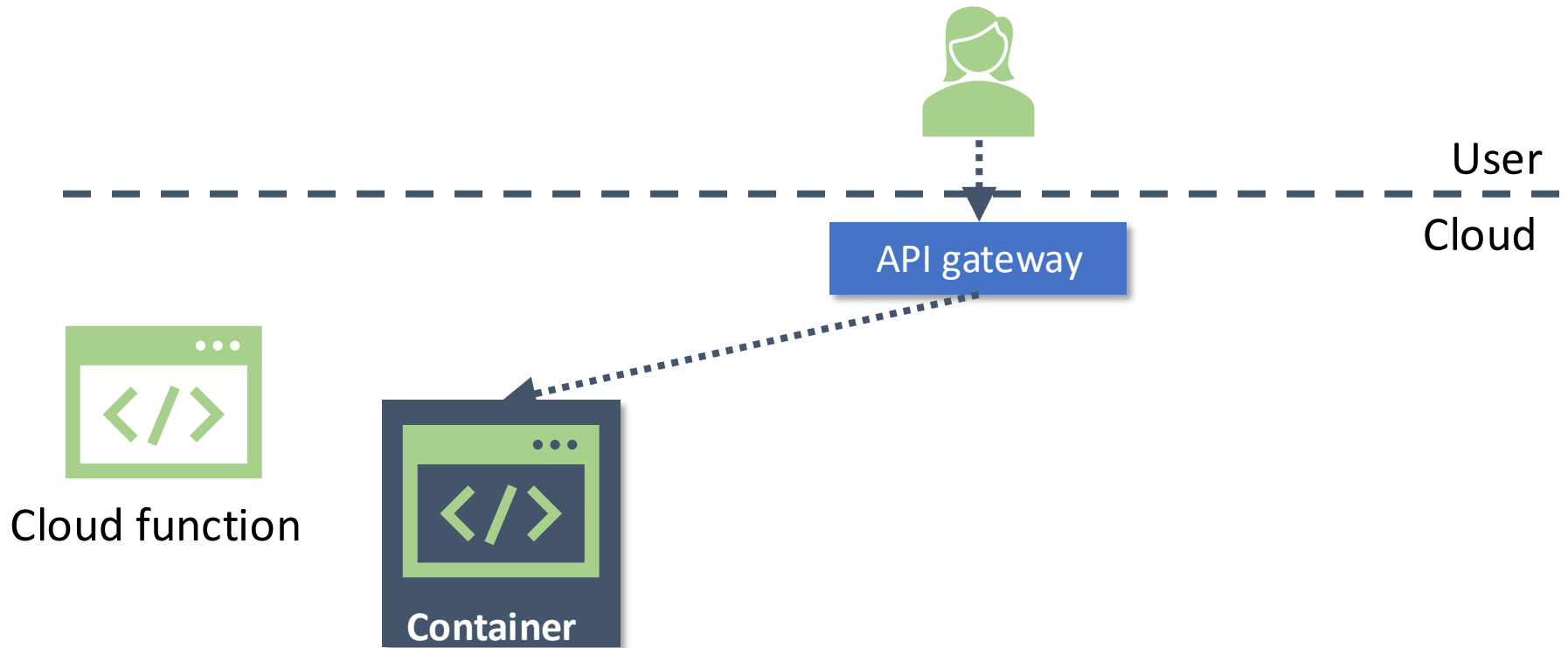
Function-as-a-Service (FaaS)



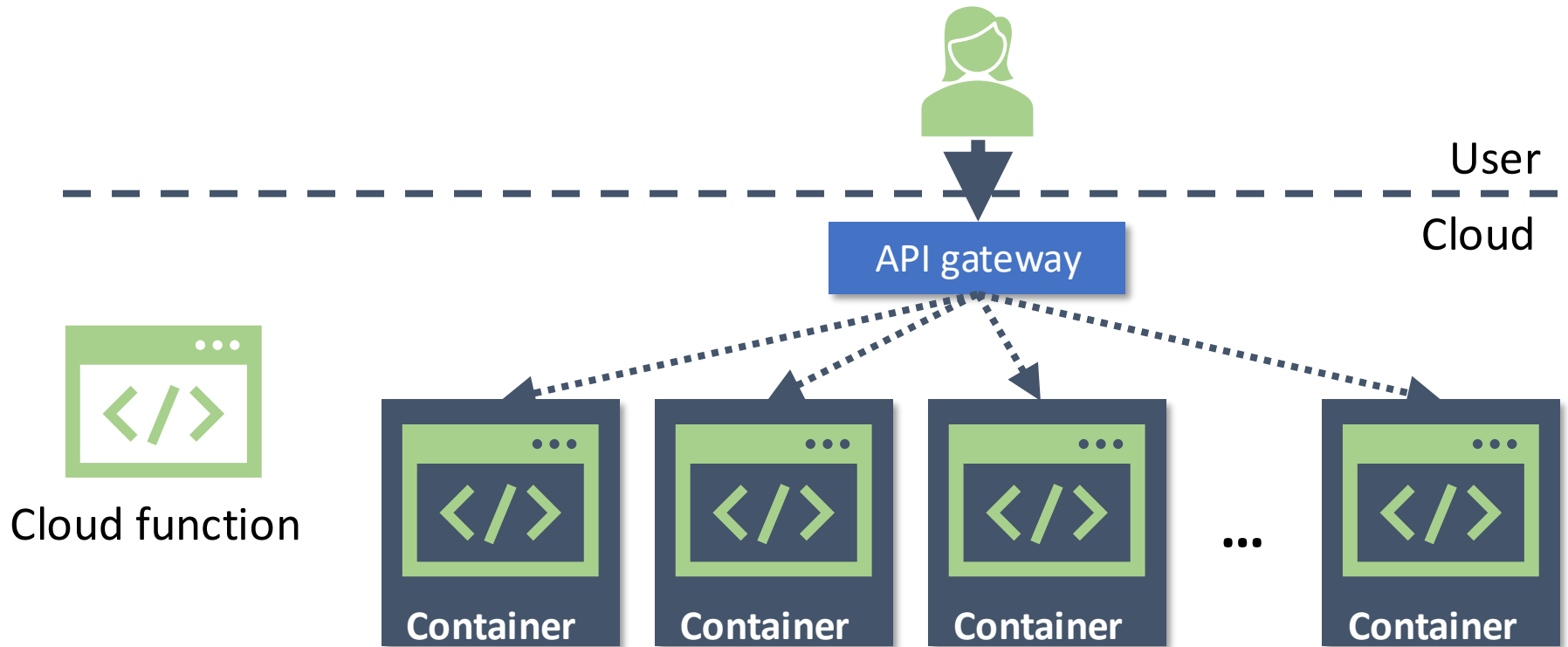
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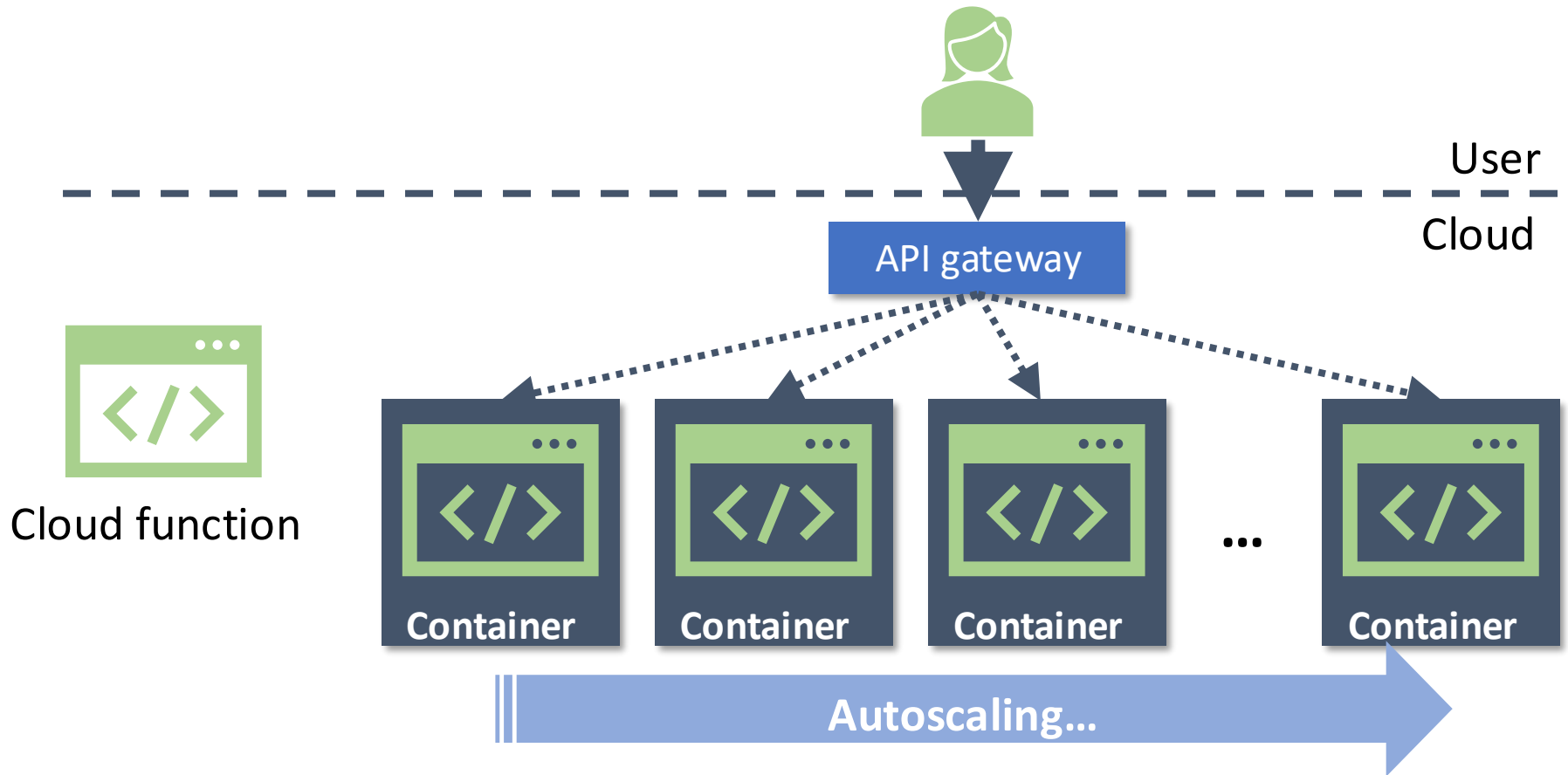
Function-as-a-Service (FaaS)



Function-as-a-Service (FaaS)



Function-as-a-Service (FaaS)



FaaS' core workflow

1. (Provider) Manage a set of user-defined functions

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6. Wait for a response

FaaS' core workflow

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4. Find an existing instance of function or create a new one
5. Send the event to the function instance
6. Wait for a response
7. Gather execution logs
8. Make the response available to the user
9. Stop the function when the execution terminates

A car analogy

Car analogy



Own a car
(Bare metal servers)



Rent a car
(VPS)



City car-sharing
(Serverless)

Cars are parked **95%** of the time (loige.link/car-parked-95)

How much do you use the car?

<https://www.slideshare.net/loige/building-a-serverless-company-with-nodejs-react-and-the-serverless-framework-jsday-2017-verona>

AWS Lambda: 1st gen → current gen



- Lambda capacity config keeps **evolving**:

1 st gen	300 seconds	900 seconds (15 minutes)	} Current offering
	single-core	two-core → up to 6 cores	
	1.5 GB	→ 10 GB memory	
	512 MB	→ up to 10GB of /tmp file system	

Python, Java, Node.js, Go, ...

Pricing:

- Fine-grained billing: **1-millisecond** billed duration
- \$0.20 per 1M requests (invocations charge \$)
- \$0.0000166667 for every GB-second (compute time charges \$\$)
- **6,000 1 GB Lambda functions for one second: 10¢**

Concept of serverless is not new

- Google App Engine
 - Fully managed platform as a service (PaaS) for developing and hosting web applications



- Google BigQuery
 - Fully managed data warehouse
 - “Arbitrarily” large data and queries
 - Pay per byte being processed
 - No concept of server or cluster



- AWS S3
 - Fully managed object storage service
 - Pay per byte being stored and written
 - No server maintenance or resource scaling



Desirable properties of FaaS

- Operationally
 - “No-ops” – (almost) no configuration
 - Autoscaling down to 0
 - Closer to pay-per-use (rather than pay-per-allocation)
 - Fine-grained billing



FaaS today

- FaaS is used mostly for simple or coarse-grained tasks
 - Stateless, embarrassingly parallel tasks, simple workflows
 - ETL, software testing, API middleware, image processing, etc.
 - Glue to other serverless backends
- Lots of **problems** are limiting FaaS' **scope**
 - Poor performance (vs. time to run actual code) at scale
 - Mismatch of infrastructure support (e.g., today's OS not designed for FaaS)
 - Rule-breaking research needed to reimagine/broaden the scope
 - Very resource-inefficient and costly for serverless providers
 - Lack of support for accelerators
 - ...
- **Orders of magnitude slower and inefficient for many “killer” apps**
 - ML inference, microservices, ...

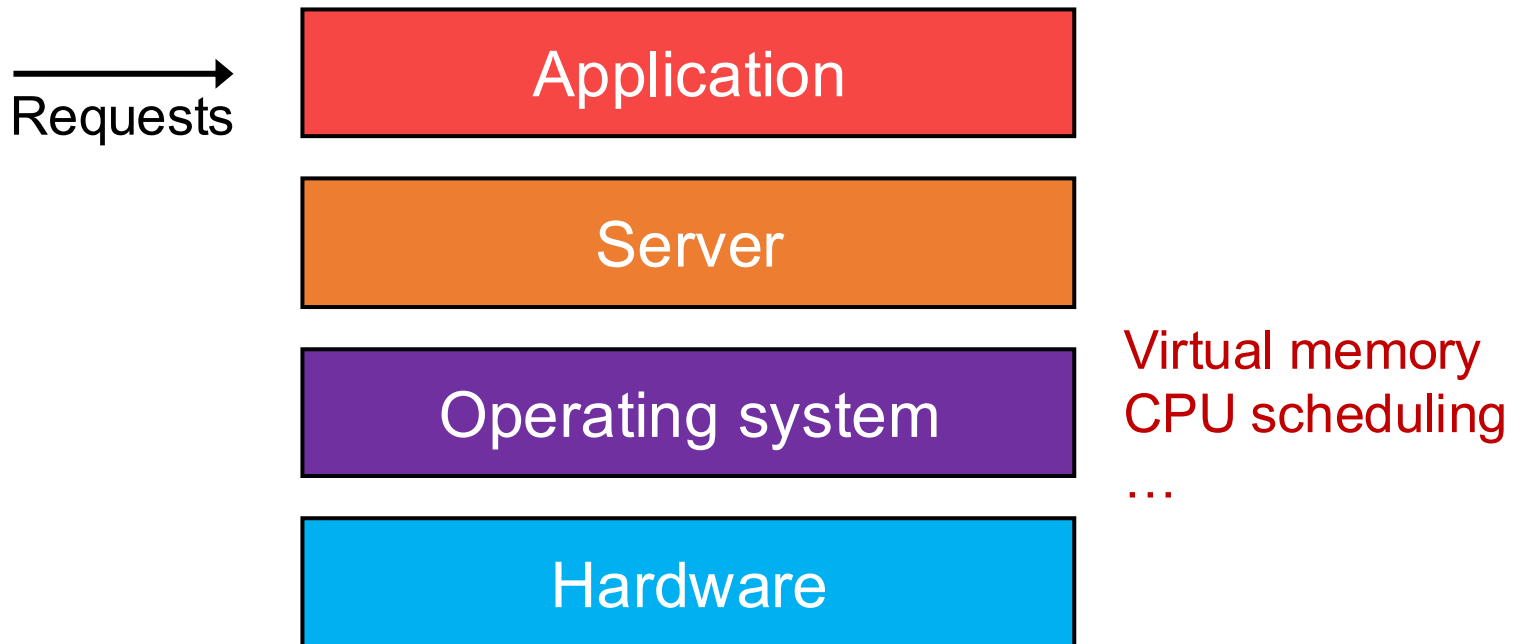
Limitations of FaaS

- **Banned** inbound network
- **No guaranteed** data availability
- Lambdas are **resource-constrained**
- Lambdas have **limited execution time**
- **High cold startup cost** and **invocation cost**

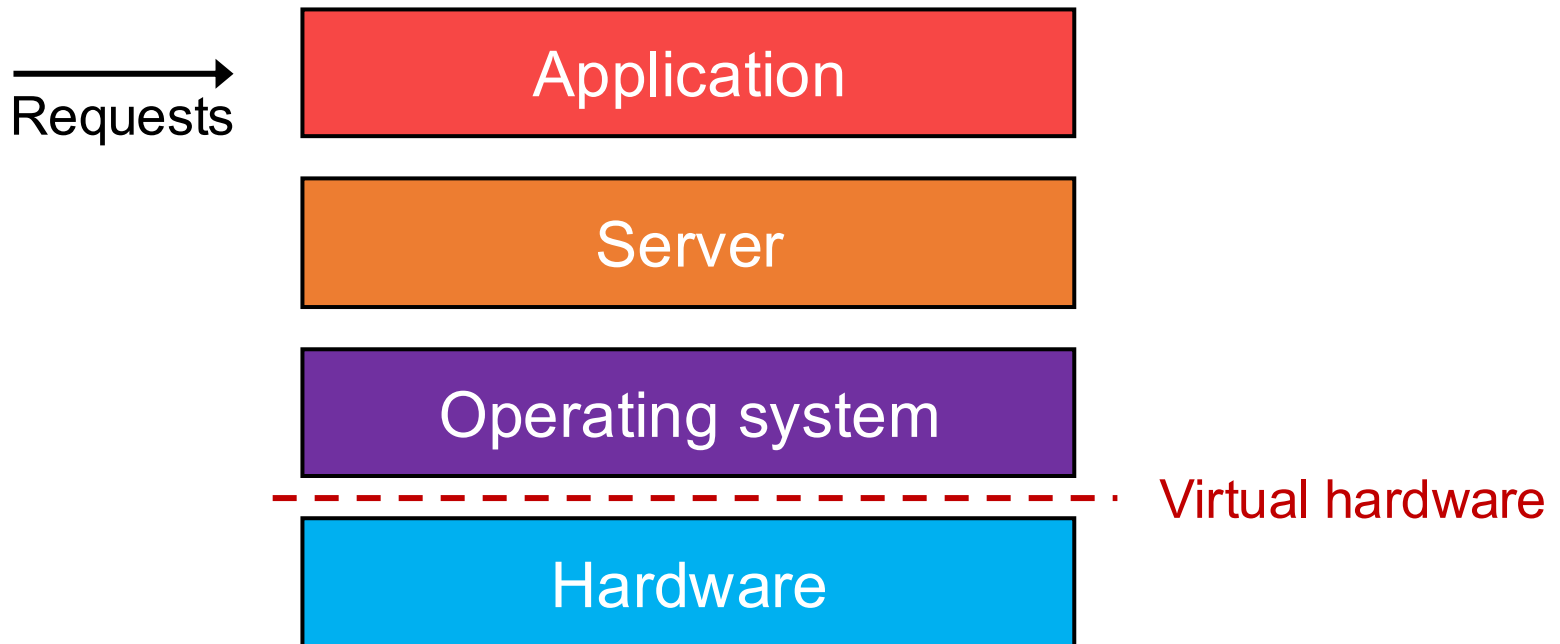
Lambda demo

Cloud evolution history – A virtualization story

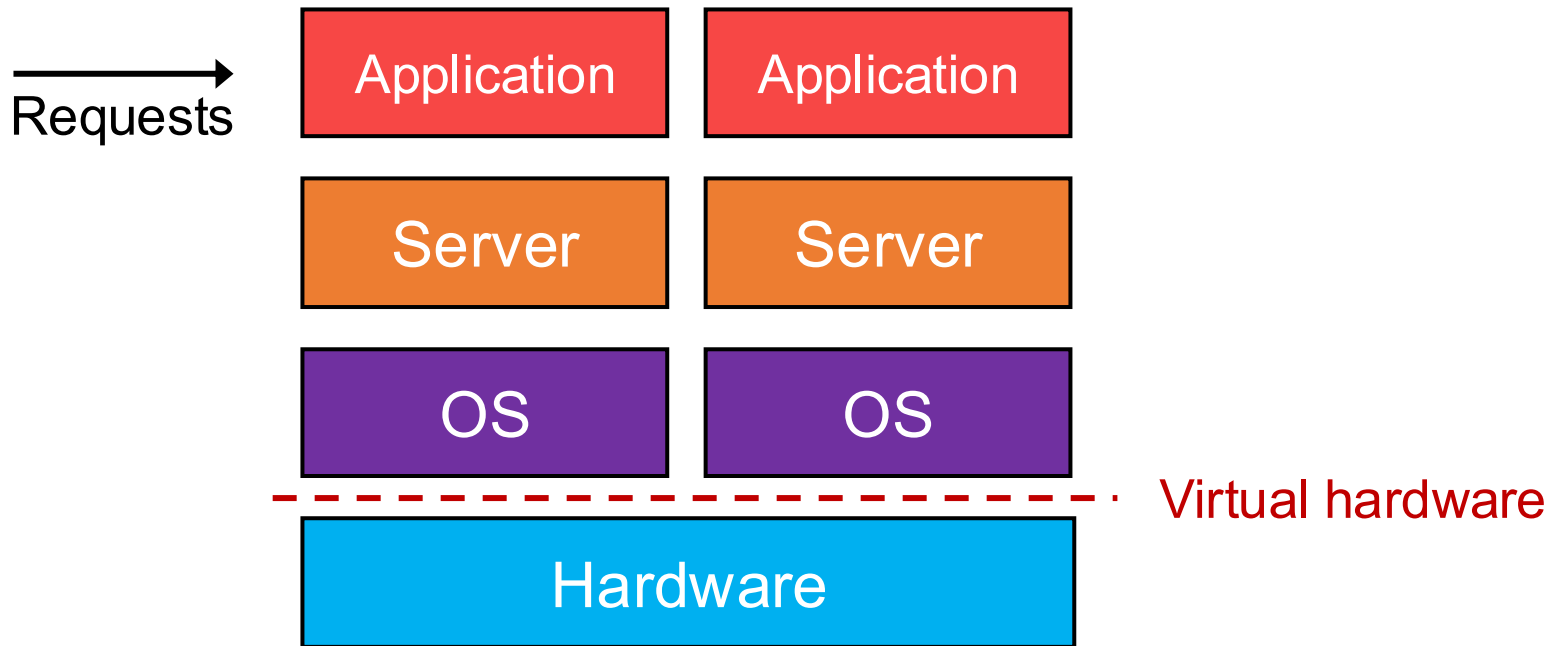
Classic server app stack



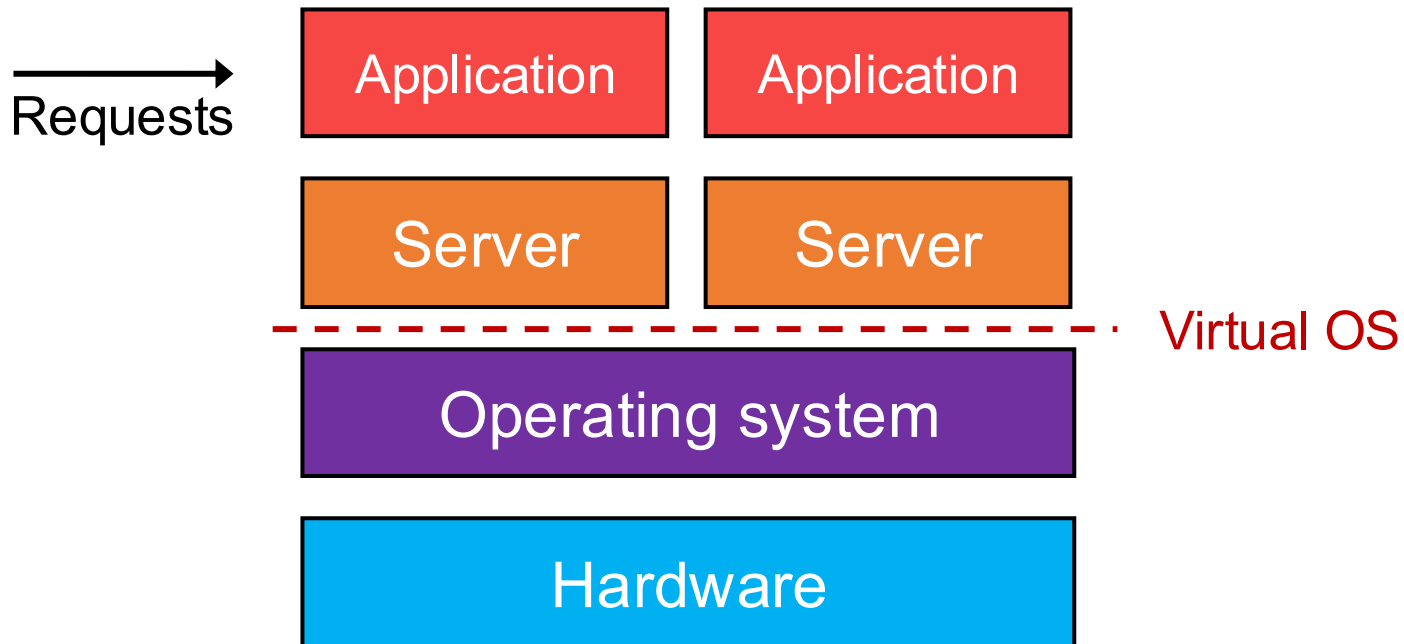
1st generation: virtual machine (VM)



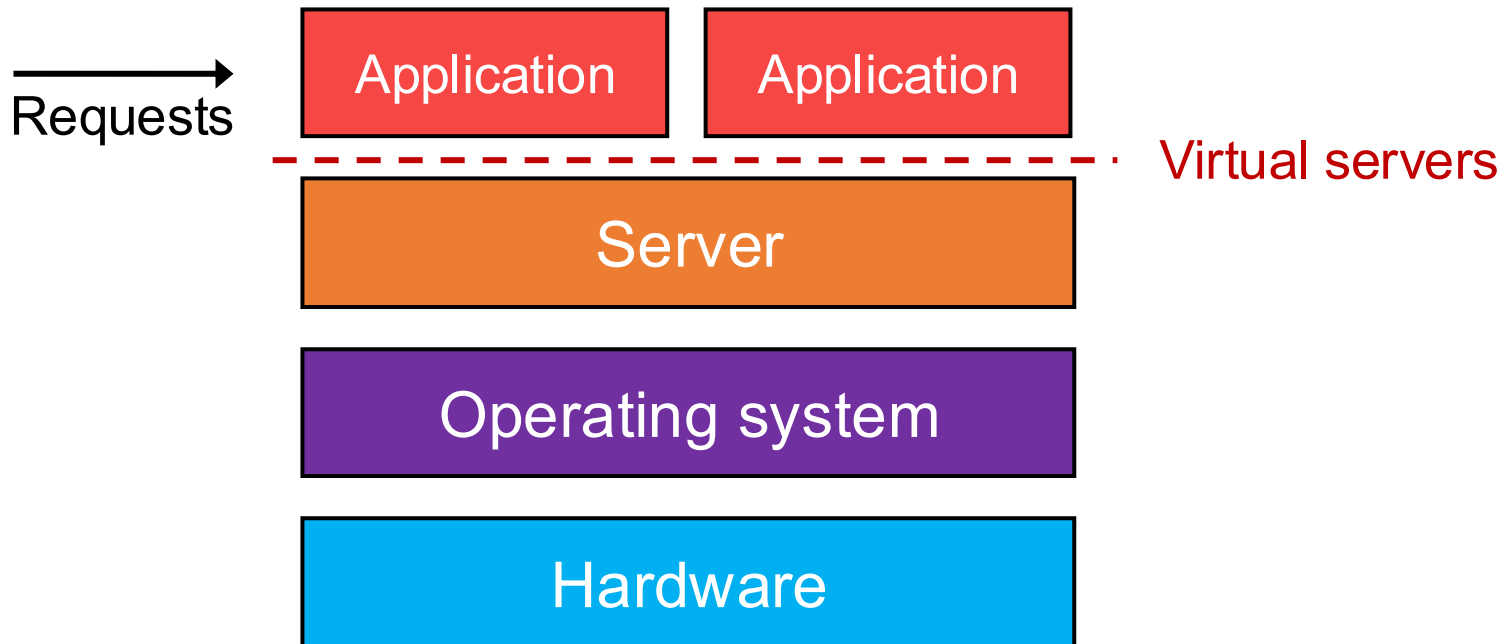
1st generation: virtual machine (VM)



2nd generation: containers



3rd generation: serverless functions



3rd generation: serverless functions

